高次元マンデルブロ集合の位相幾何学的性質について

19 Feb 2025

談話会@京大人環

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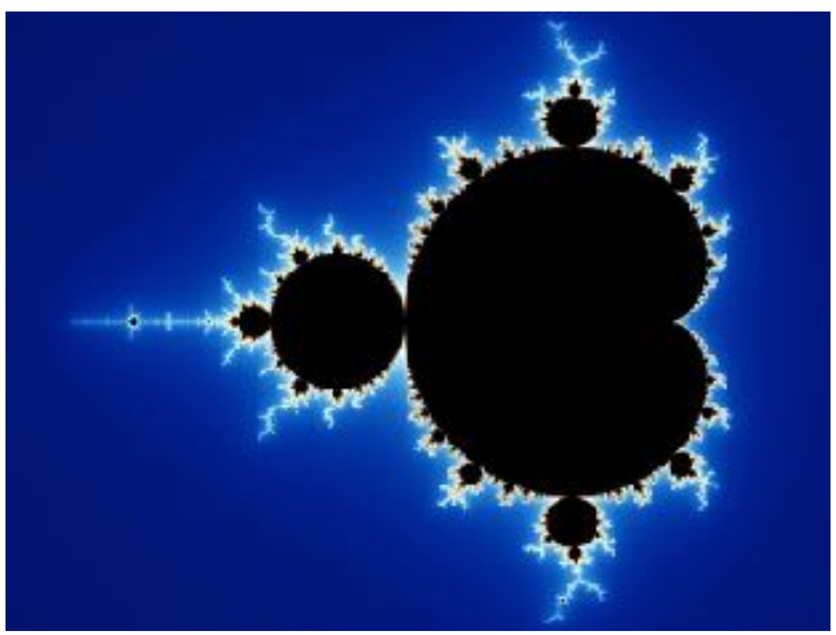
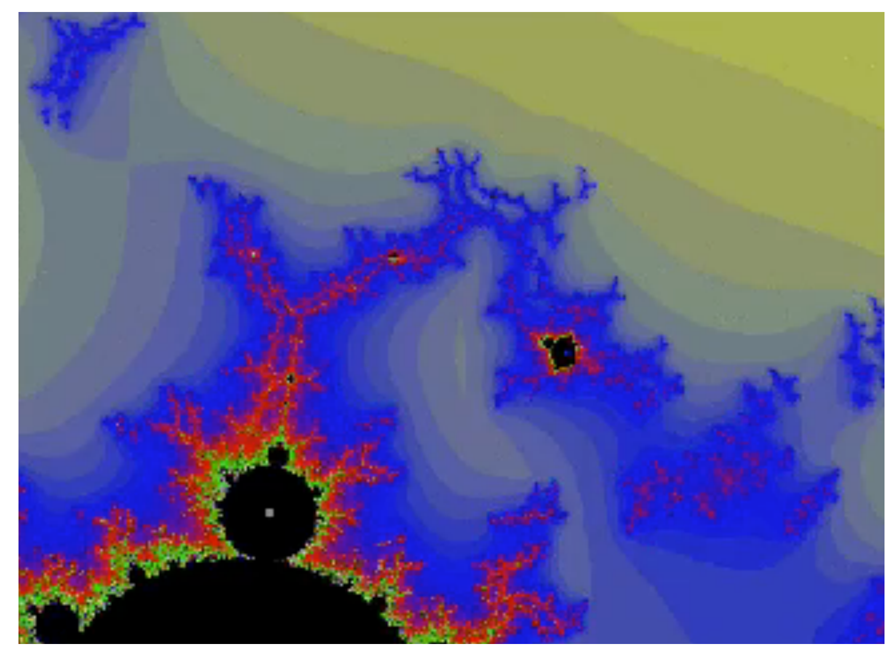
Contents

Aim: We want to understand the topology of the higher dimensional analogue of Mandelbrot set.





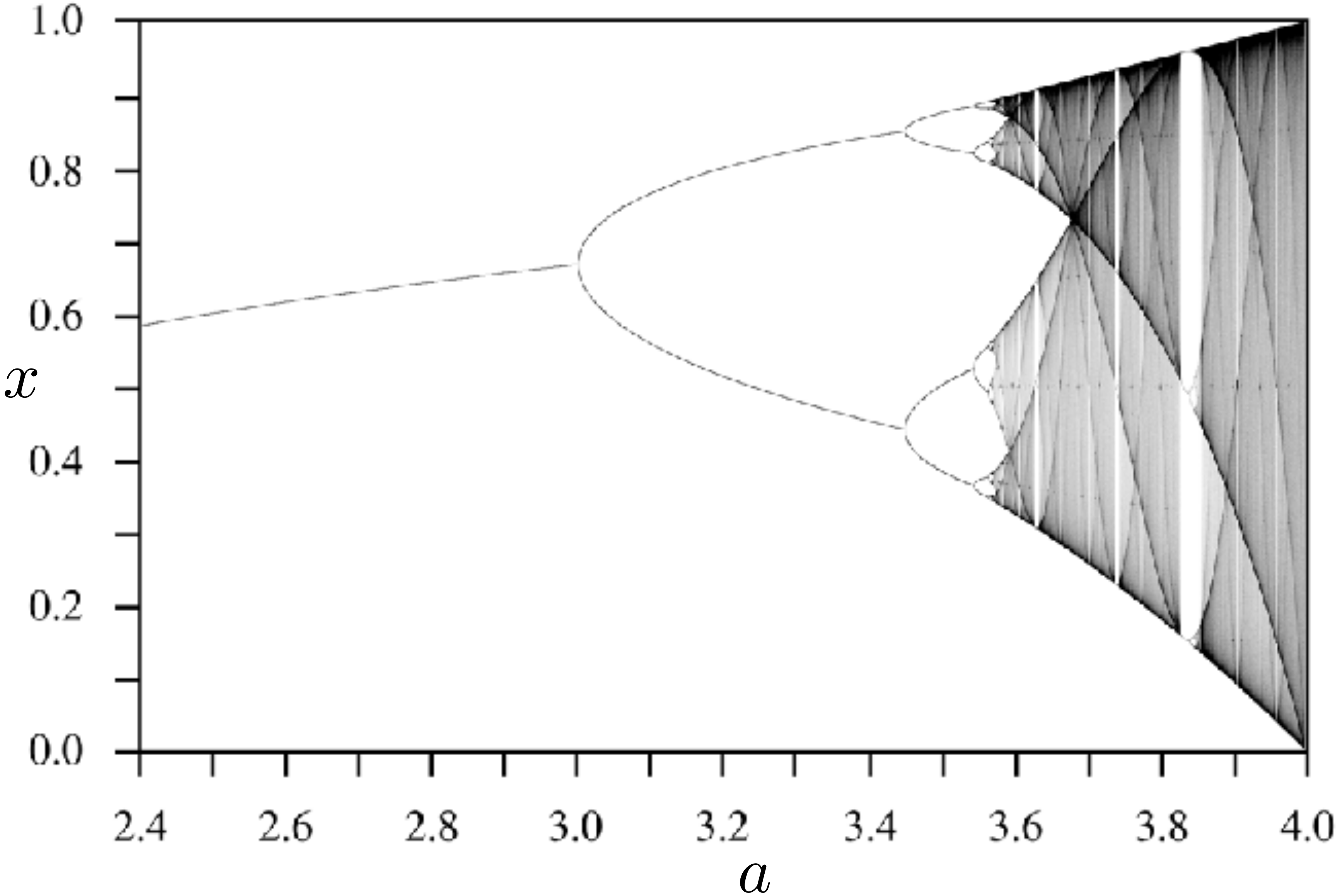
1. Introduction
2. Monodromy representation
3. Connectedness and Disconnectedness

Introduction (1):
1 dim Quadratic Map

Quadratic Map and Period Doubling

$$f_a = ax(1-x)$$

$$f_a^n := f_a \circ f_a \circ \cdots \circ f_a \ \text{(n-times)}$$



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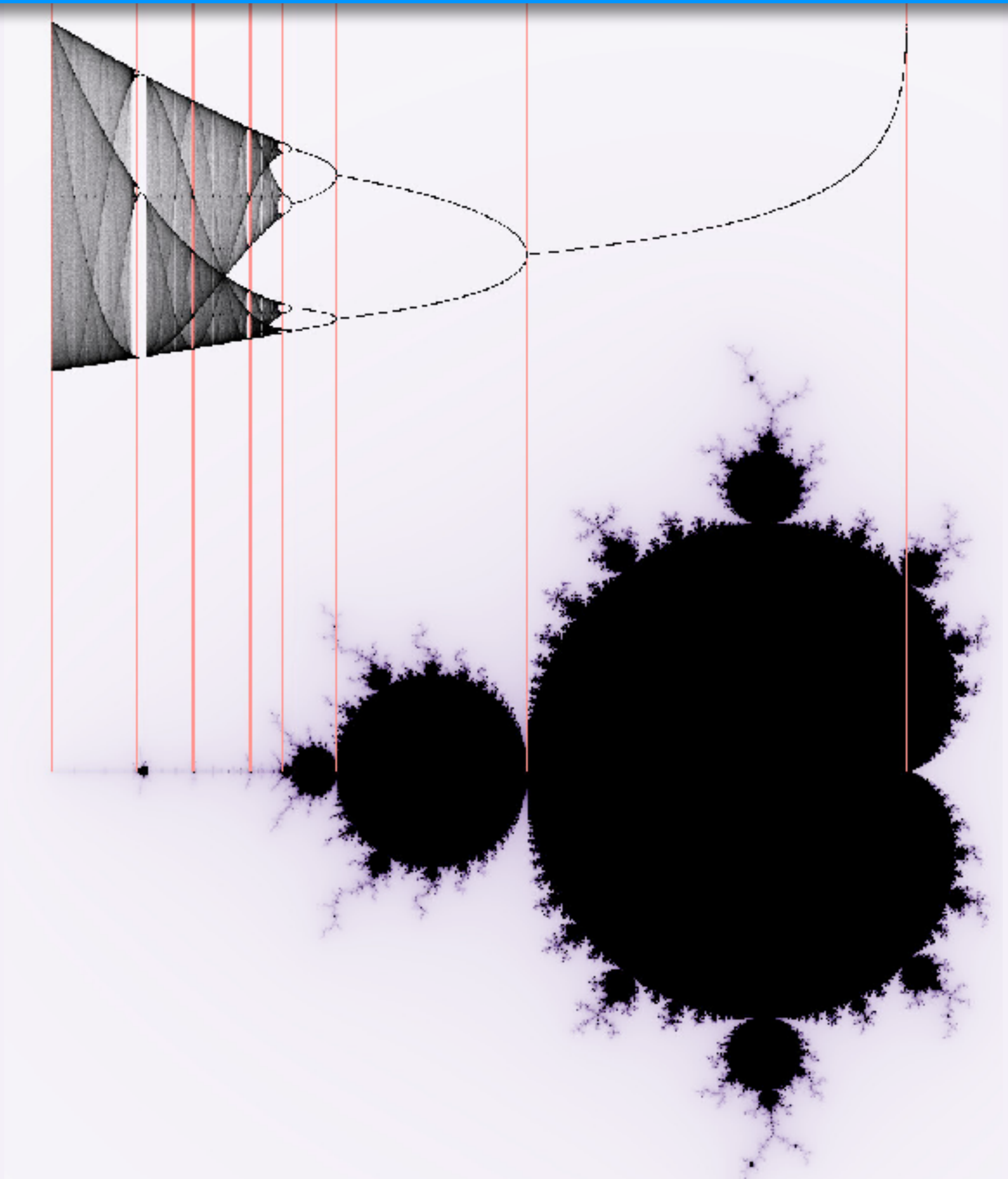
Real and Complex Bifurcations



The Complex Quadratic Map

The complex quadratic map

$$P_c : \mathbb{C} \to \mathbb{C} : z \mapsto z^2 + c \qquad (c \in \mathbb{C} \text{ is the parameter})$$

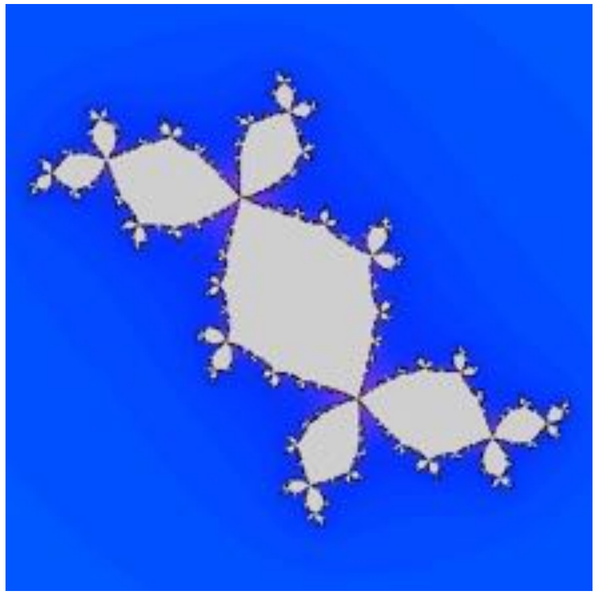
The simplest nonlinear dynamics

$$P_c^n := P_c \circ P_c \circ \cdots \circ P_c \text{ (n-times)}$$

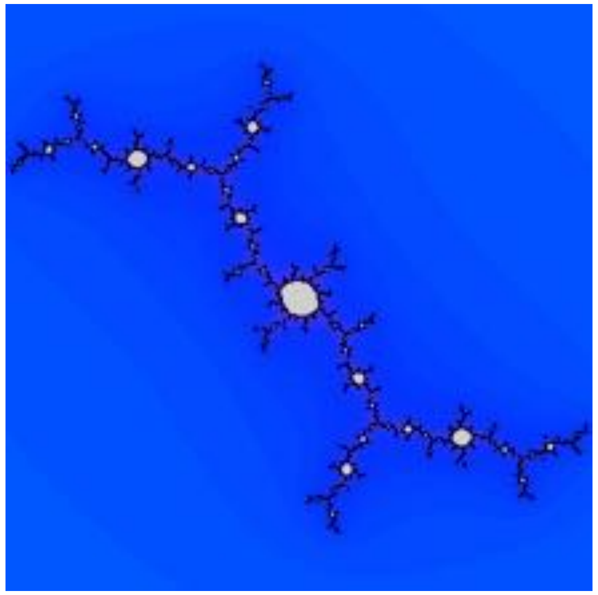
Gaston Julia

The filled Julia set and the Julia set

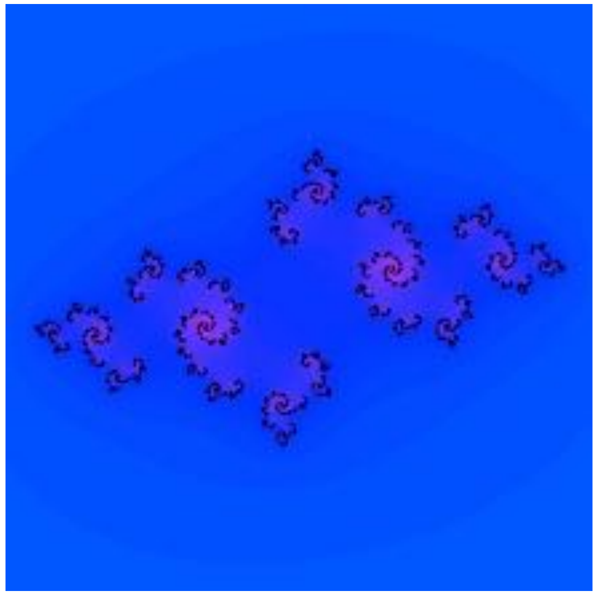
$$K_c := \{z \in \mathbb{C} \mid \{P_c^n(z)\} \text{ is bounded}\} \qquad J_c := \partial K_c$$



$c = -0.125475 + 0.7494i$



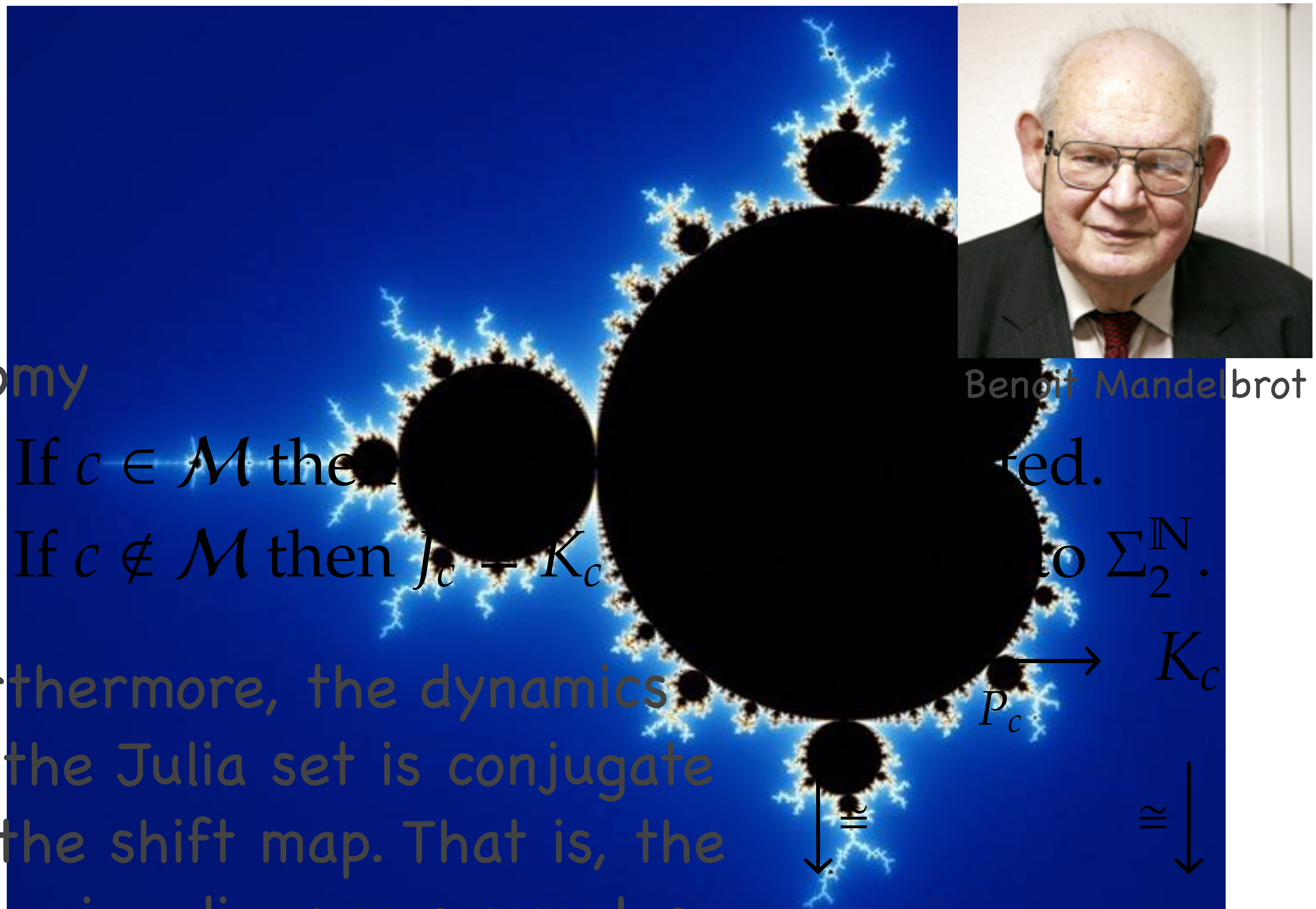
$c = -0.157414272428 + 1.033054947853i$



$c = -0.810000061989 - 0.344999969006i$

Mandelbrot Set

The Mandelbrot Set $\mathcal{M} := \{c \in \mathbb{C} \mid \{P_c^n(0)\} \text{ is bounded}\}$



Benoit Mandelbrot

Dichotomy

$A.$ If $c \in \mathcal{M}$ the ted.

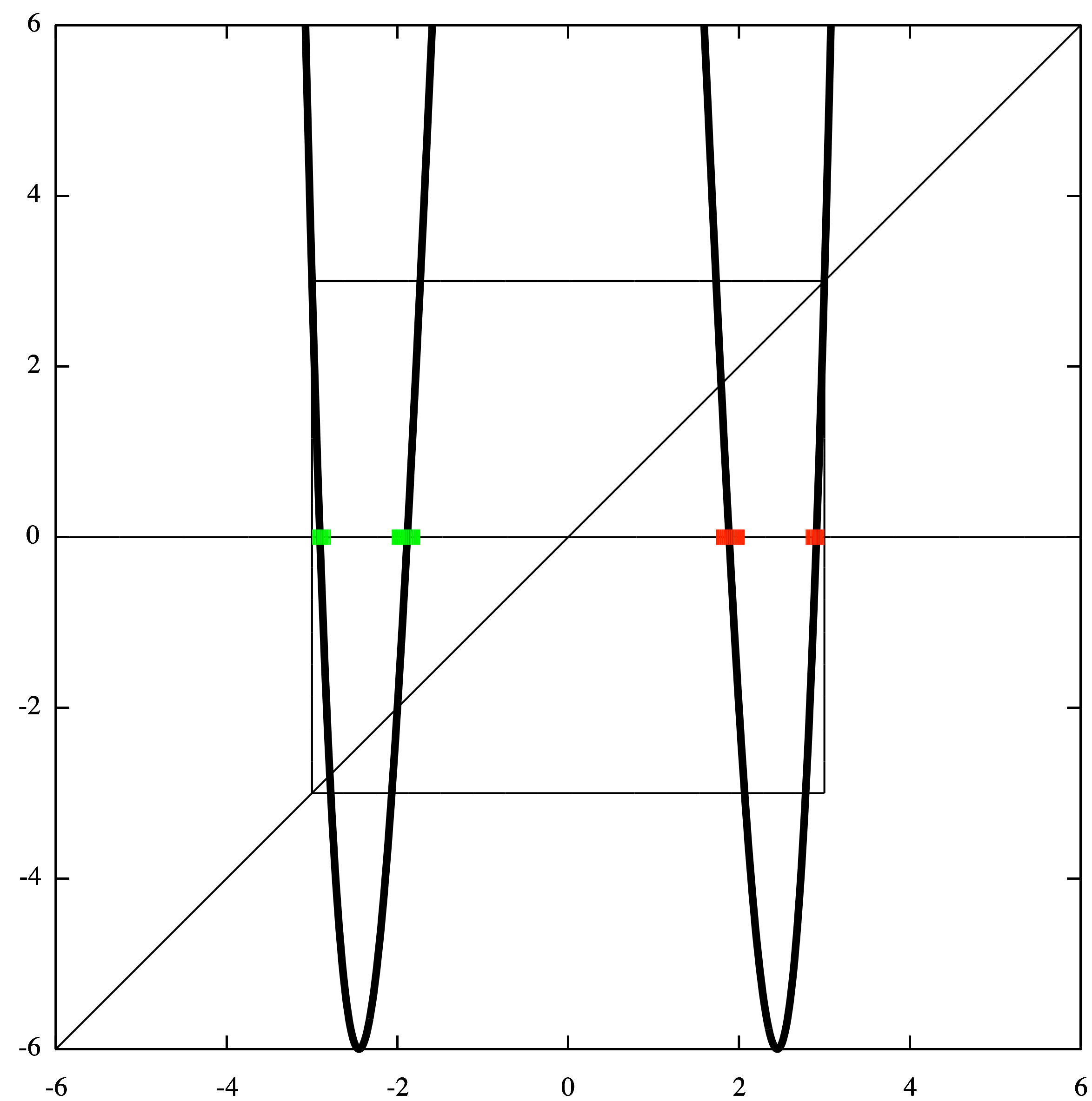
$B.$ If $c \notin \mathcal{M}$ then $J_c = K_c$ to $\Sigma_2^{\mathbb{N}}$.

Furthermore, the dynamics on the Julia set is conjugate to the shift map. That is, the following diagram commutes.

$$\begin{tikzcd} {} \arrow[r, "P_c"] \arrow[d, "\cong"] & K_c \arrow[d, "\cong"] \\ \Sigma_2^N \arrow[r, "\sigma"] & \Sigma_2^{\mathbb{N}} \end{tikzcd}$$

Cantor Julia Set

Assume $c = -6$ then $\mathbb{R} \subset \mathbb{C}$ is invariant.



One-sided shift space

one-sided shift space

$\{0, 1, \ldots, d-1\}$: alphabet

$\Sigma_d^{\mathbb{N}} := \{0, 1, \ldots, d-1\}^{\mathbb{N}}$: one-sided shift space

metric on the shift space

For $s = (s_0\ s_1\ s_2\ \cdots)$ and $t = (t_0\ t_1\ t_2 \cdots)$,

$$d(s,t) := \sum_{i=0}^{\infty} \frac{1-\delta(s_i, t_i)}{2^i} \qquad (\delta \text{ is the Kronecker delta })$$

Two sequences are close provided their first few entries agree.

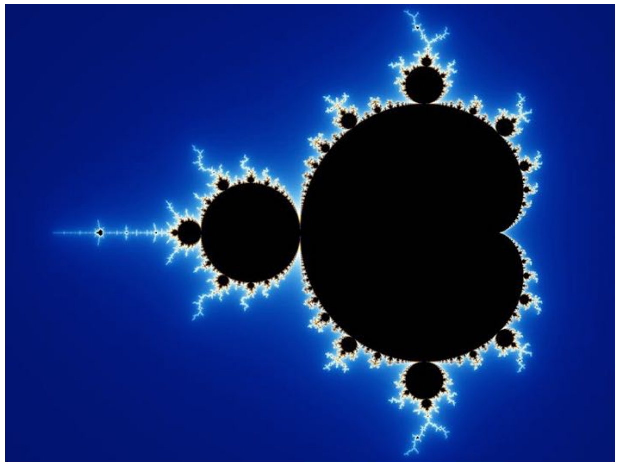
one-sided shift map

$$\sigma : \Sigma_d^{\mathbb{N}} \to \Sigma_d^{\mathbb{N}} : (s_0\ s_1\ s_2 \cdots) \mapsto (s_1\ s_2\ s_3 \cdots).$$

Continuous, d-to-1 map

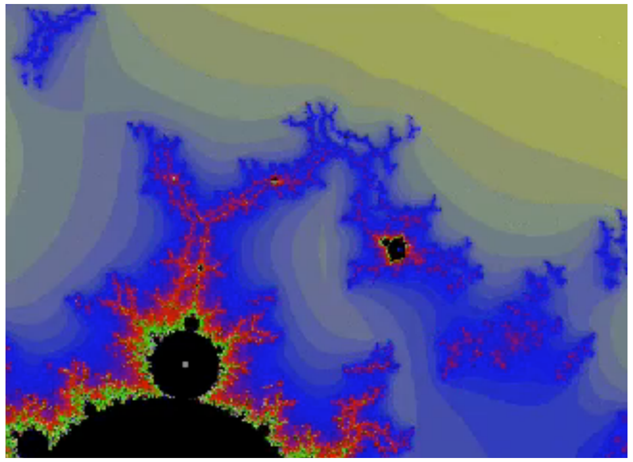
Mandelbrot Set

- $\mathscr{M}$ is connected
- $\mathscr{M}^c$ is isomorphic to $\mathbb{C} \backslash \mathbb{D}$
- The Hausdorff dim of $\partial \mathscr{M}$ is 2
- It is conjectured that $\mathscr{M}$ is locally connected



Today: We will consider $\mathscr{M} \subset \mathbb{C}^2$, a generalized version of the Mandelbrot set, and show that

- $\pi_1(\mathscr{M}^c)$ is much larger than $\mathbb{Z}$
- $\mathscr{M} \cap \mathbb{R}^2$ is NOT connected



Introduction (2):
Hénon Map

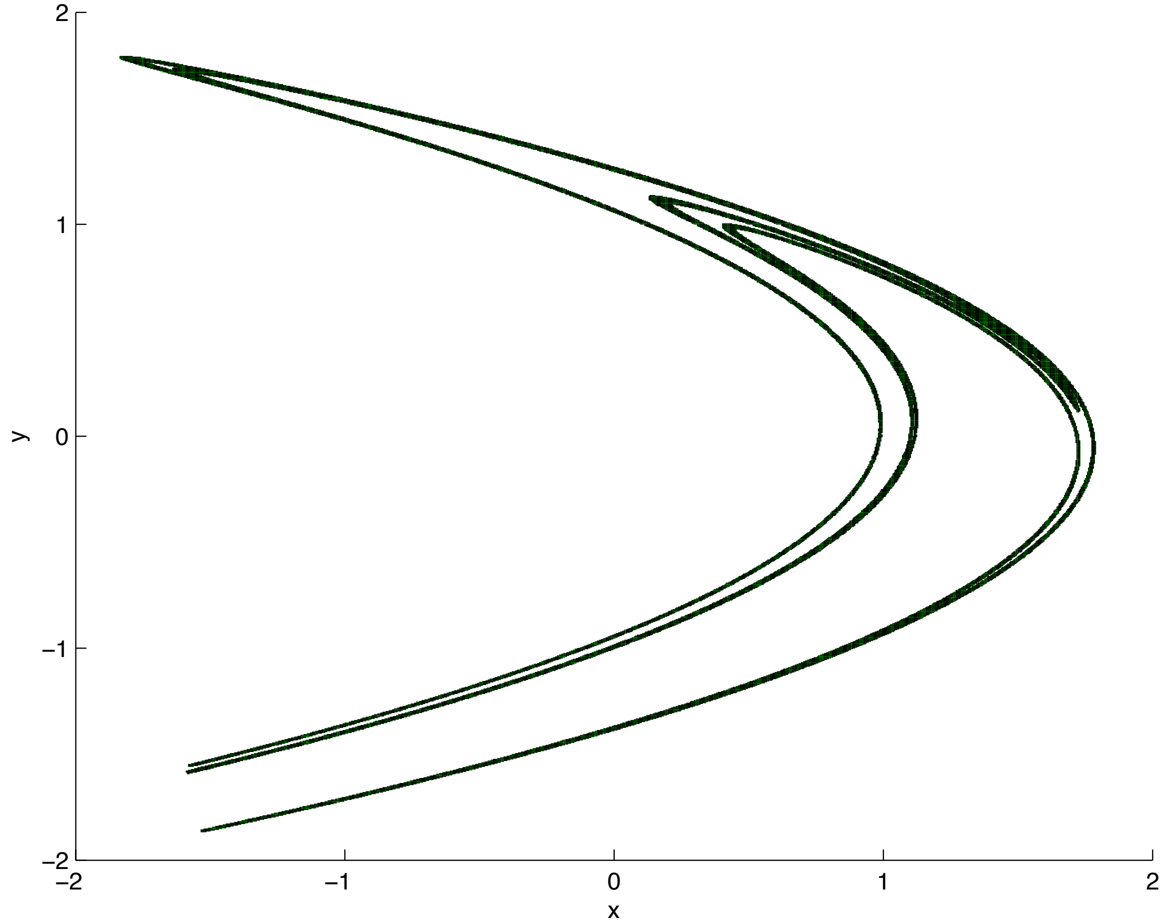
The Hénon Map

The *Hénon family* on $\mathbb{R}^2$:

$$f_{a,b} : \mathbb{R}^2 \ni (x, y) \longmapsto (x^2 - a - by, x) \in \mathbb{R}^2,$$

where $(a, b) \in \mathbb{R} \times \mathbb{R}^\times$.

Michel Hénon
1931-2013



strange attractor for $(a, b) = (1.4, -0.3)$.

The Hénon map

Hénon map is:

- a diffeomorphism if $b \neq 0$.
- conjugate to the quadratic map if $b = 0$.
- only nontrivial 2-dim polynomial diffeomorphism of degree 2.
- a role-model of nonlinear dynamics in dimension higher than 2.

Michel Hénon
1931–2013

If $|b|$ is sufficiently large, the Julia set of the Hénon map is conjugate to the shift map of the two-sided shift space of two symbols $\Sigma_2^{\mathbb{Z}}$.

Two-sided shift space

two-sided shift space

$\{0, 1, \ldots, d-1\}$: alphabet

$\Sigma_2^{\mathbb{Z}} = \{0, 1, \ldots, d-1\}^{\mathbb{Z}}$: two-sided shift space

metric on the shift space

For $s = (\cdots s_{-2}\, s_{-1} \,.\, s_0\, s_1 \cdots)$ and $s = (\cdots t_{-2}\, t_{-1} \,.\, t_0\, t_1 \cdots)$,

$$d(s,t) := \sum_{-\infty}^{\infty} \frac{1-\delta(s_i, t_i)}{2^{|i|}} \qquad (\delta \text{ is the Kronecker delta })$$

Two sequences are close if their entries in the middle agree.

two-sided shift map

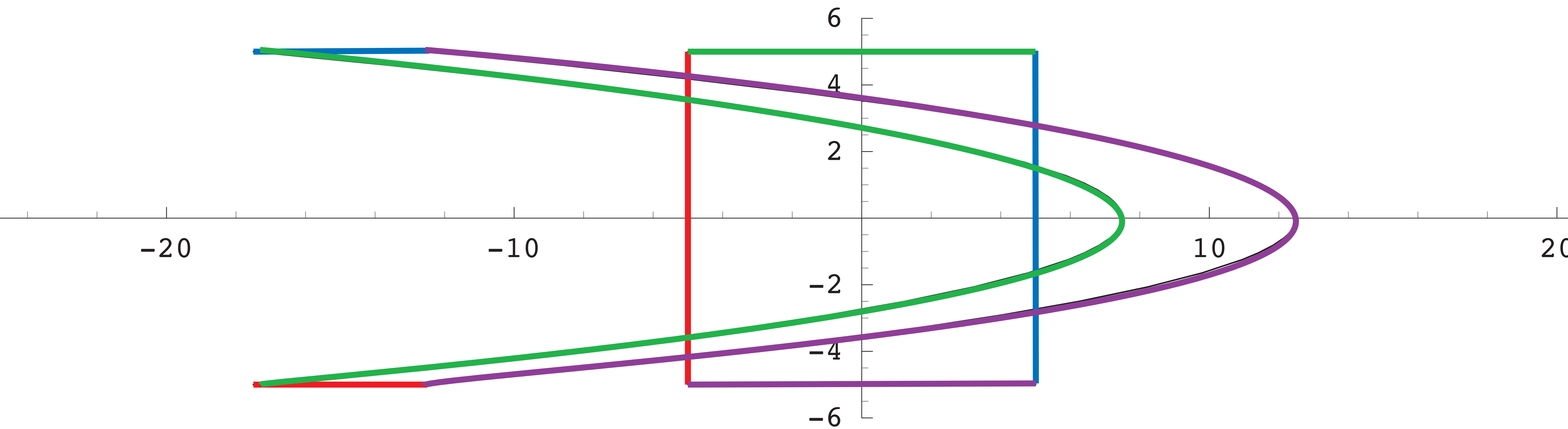
$$\sigma : \Sigma_d^{\mathbb{Z}} \to \Sigma_d^{\mathbb{Z}} : (\cdots s_{-2}\, s_{-1} \,.\, s_0\, s_1 \cdots) \mapsto (\cdots s_{-1}\, s_0 \,.\, s_1\, s_2 \cdots)$$

homeomorphism

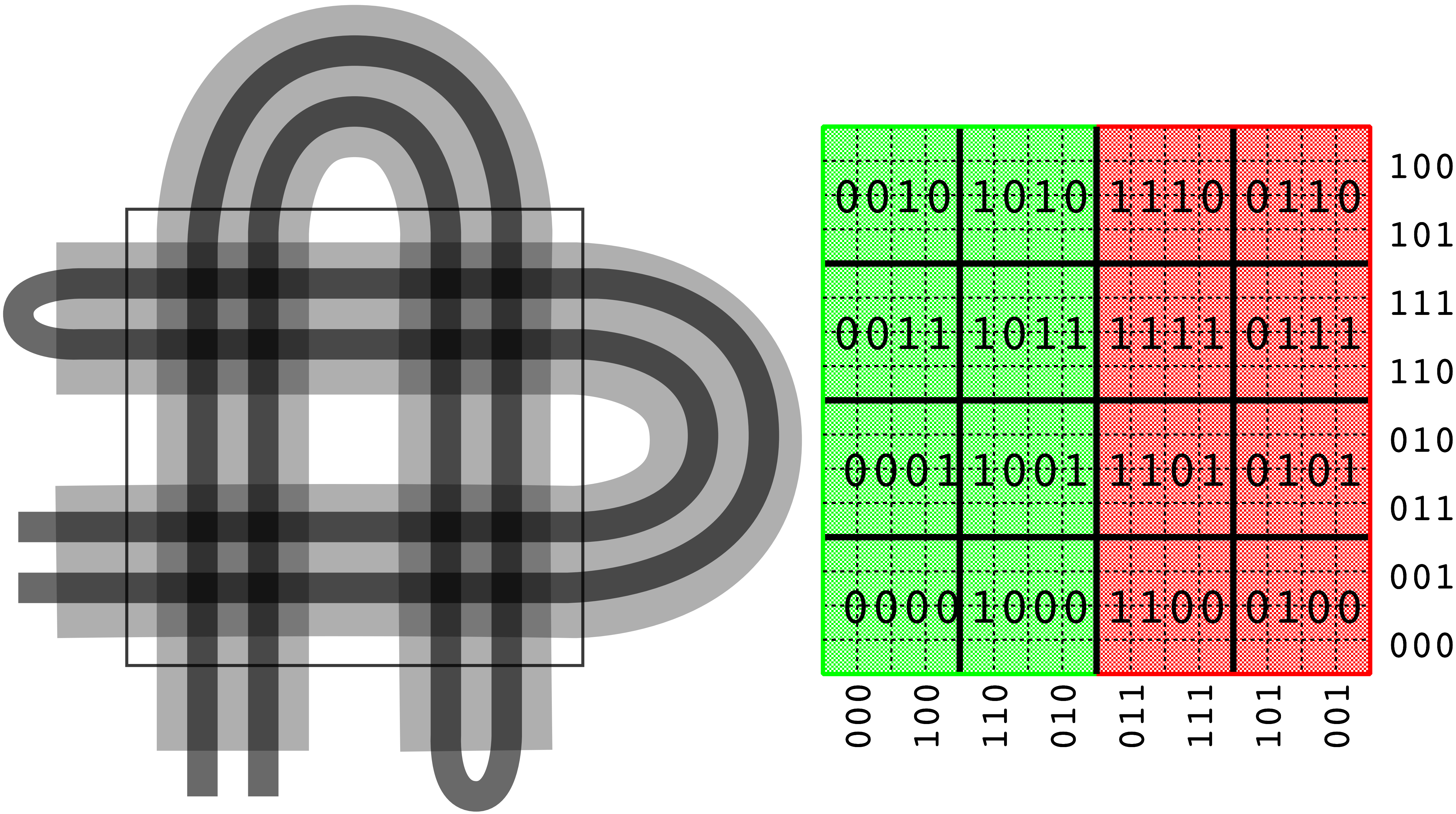
Horseshoe in the Hénon

$$(x, y) \mapsto (a - x^2 + by, x)$$

[Devaney and Nitecki 1979] For any fixed b, if a is sufficiently large, then the non-wandering set of the Hénon map is uniformly hyperbolic full horseshoe.

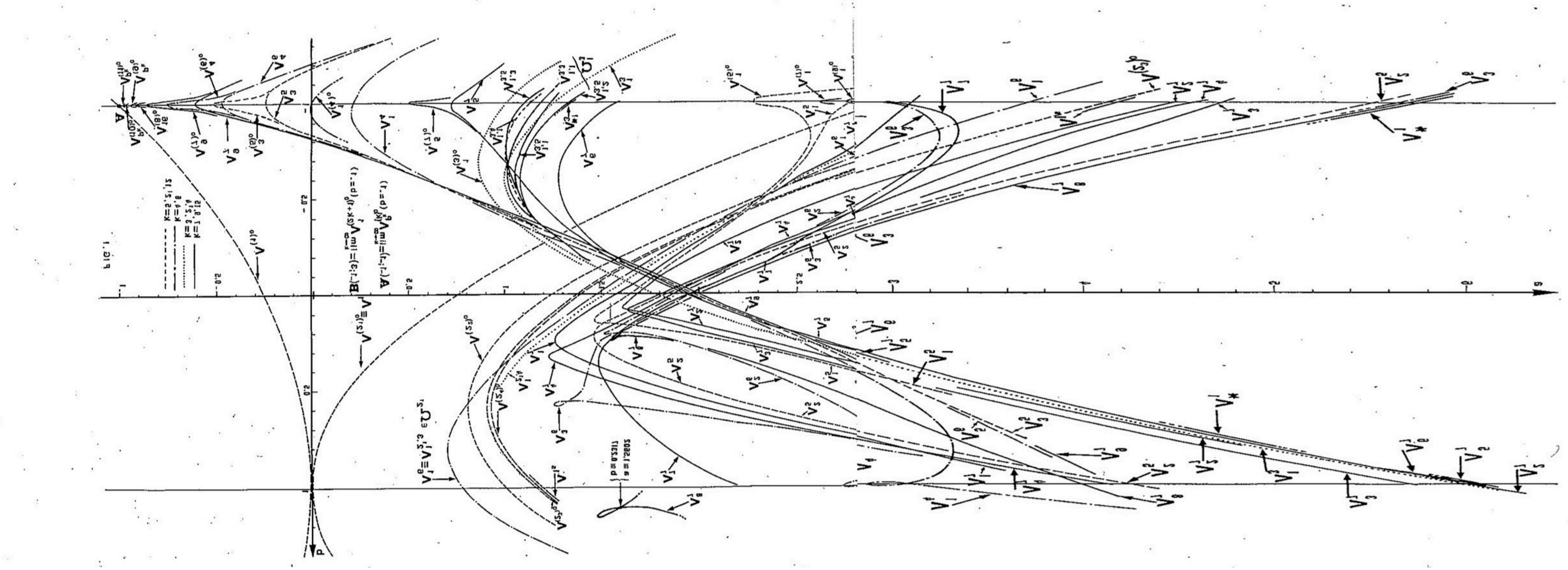


Symbolic Coding of Horseshoe

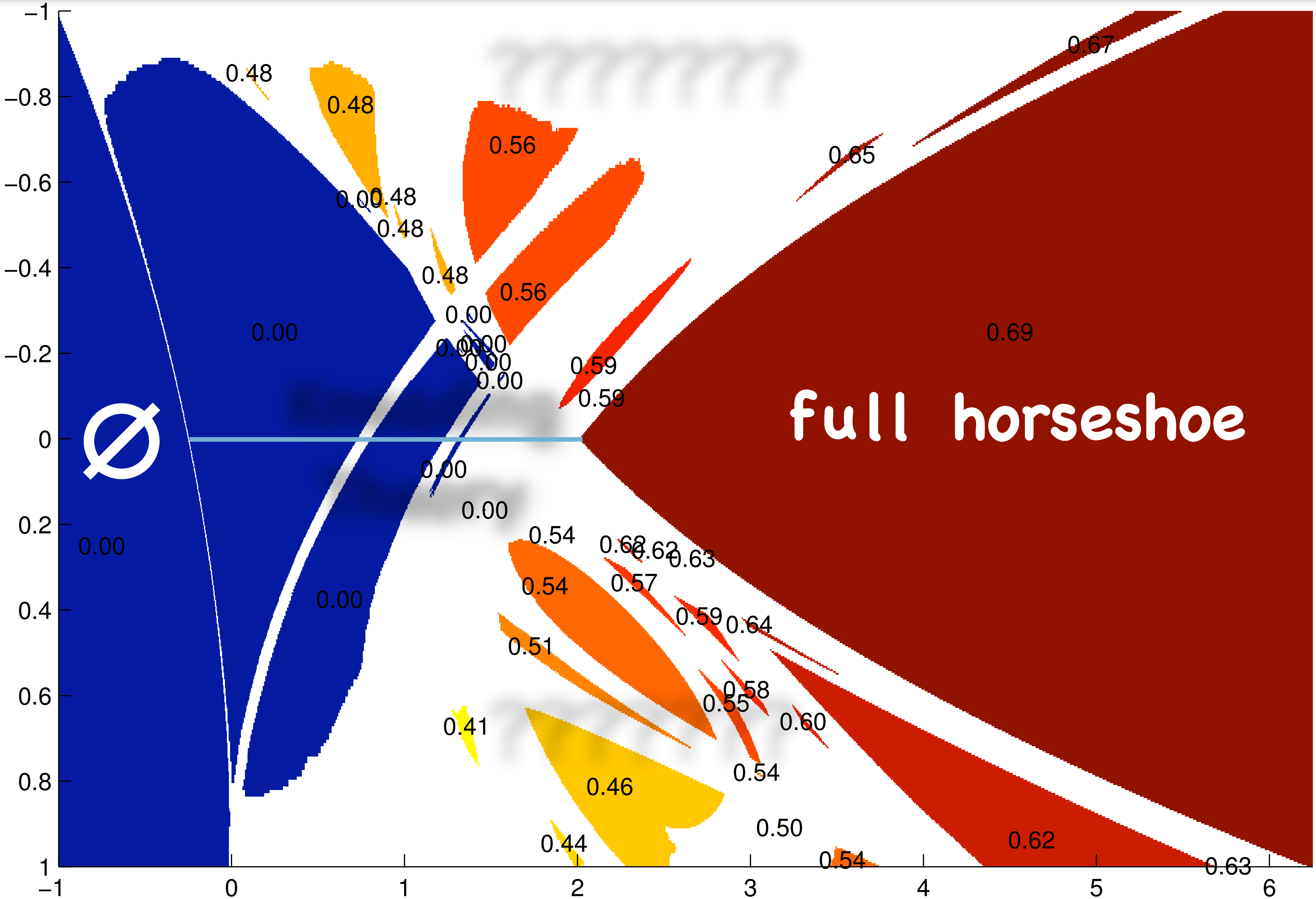


Numerical Bifurcation Curves

A figure by Mira and El Hamouly (C.R.A.S.Paris, 1981).



Creation of horseshoe



Hyperbolic plateaus of the real Hénon map $(x, y) \mapsto (a - x^2 + by, x)$
and rigorous lower bounds of the topological entropy

Kneading Theory

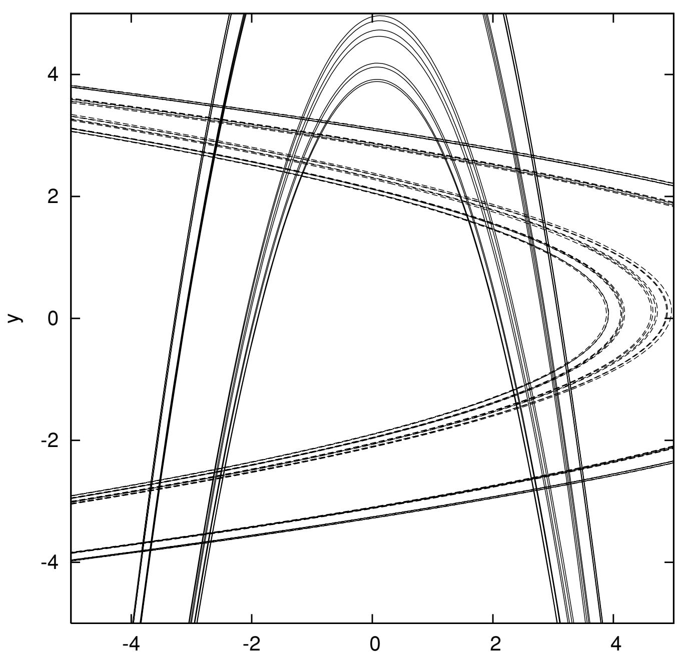
For 1-dim real unimodal maps, we already have almost complete understanding of the creation of chaos:
"Milnor-Thurston's Kneading Theory"

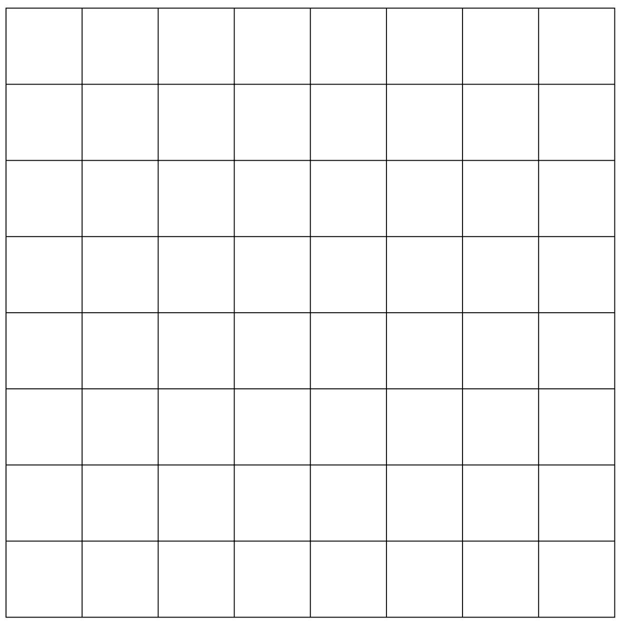
* To each point in the interval we can assign a symbolic coding called "kneading coordinate".
* We can regard the dynamics as a subshift.
* The subshift is determined by the kneading coordinate of he critical point through the "kneading order".
* Topological entropy is monotone increasing.

Pruning Front

Phase Space

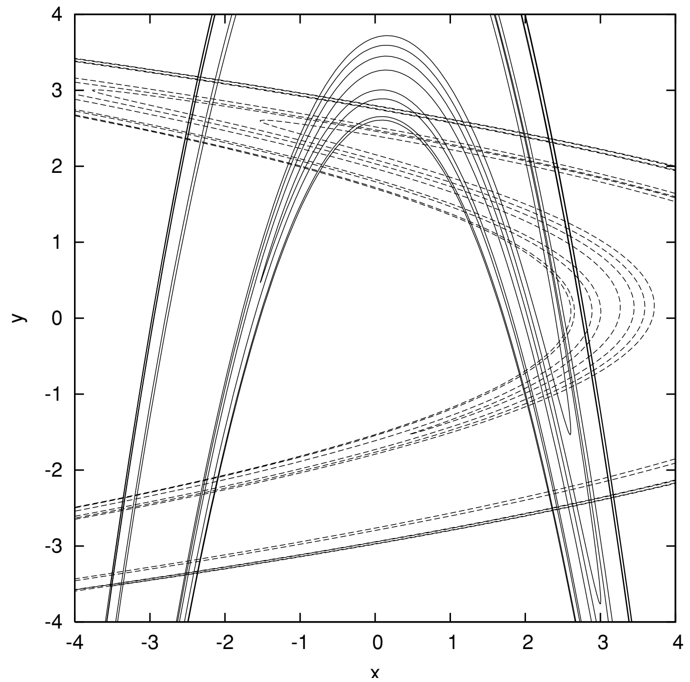
"Symbolic Plane"





Complete Horseshoe

There is a 1-to-1 correspondence



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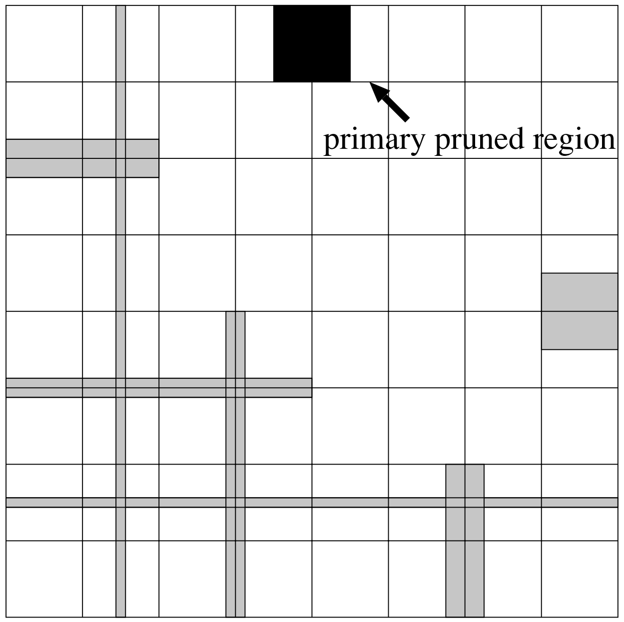


Incomplete Horseshoe?

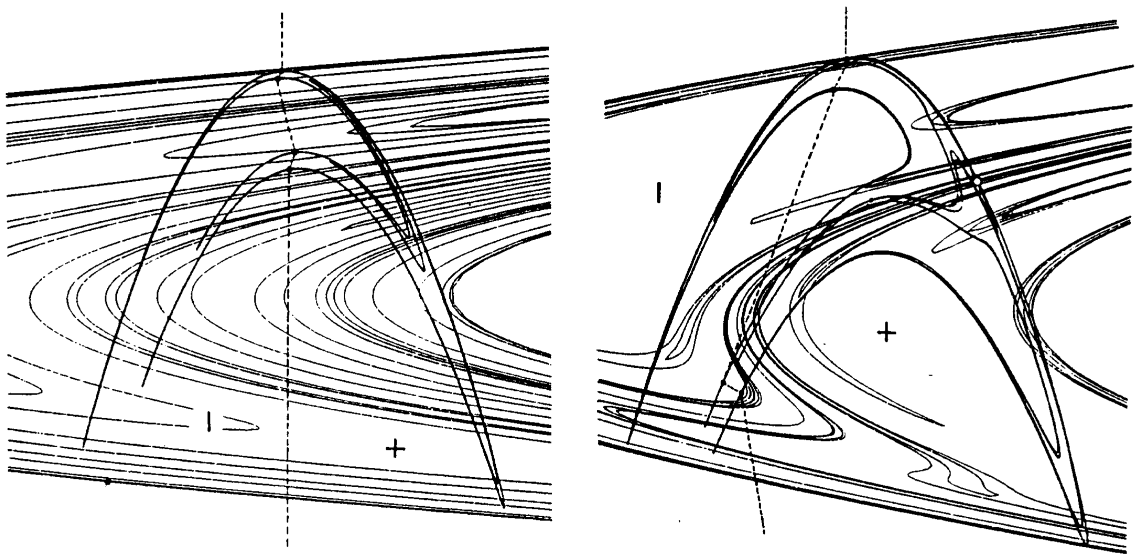
Some Symbols are "pruned" away...

Figures from Hagiwara-Shudo, J. Phys. A: Math. Gen. 37 (2004)

Pruning Front Conjecture


Predrag Cvitanović

Conjecture (P. Cvitanović)
The dynamics of the real Hénon map can be described as an "incomplete horseshoe" defined by a finite pruning front.



Difficulties...

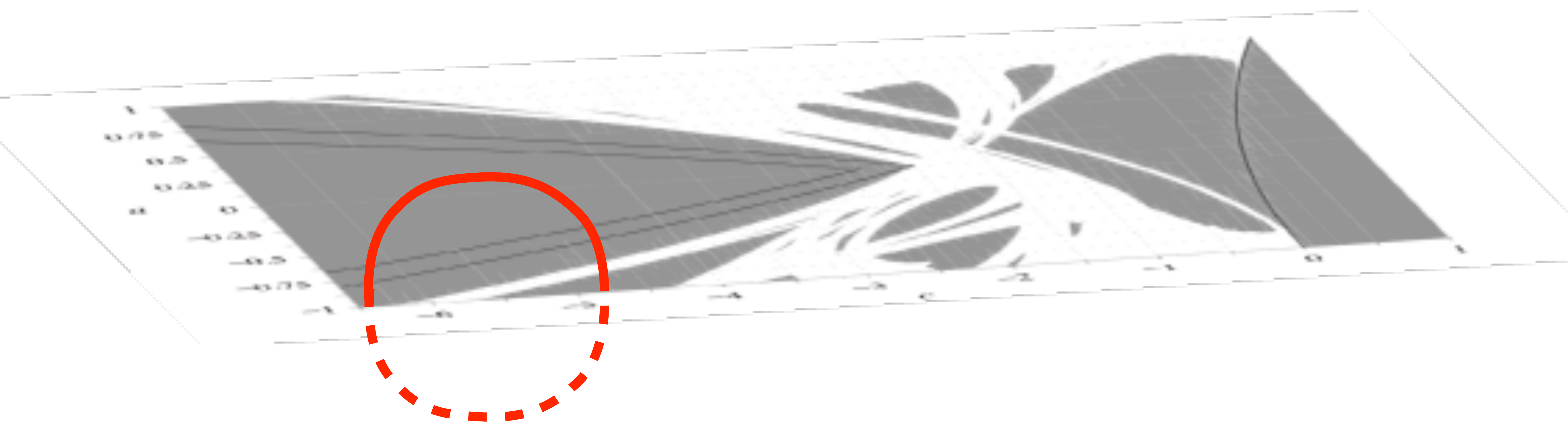
* There exist counter examples.
* No canonical symbolic coding.

Our Strategy

Now we consider the complex Hénon map:

$$H_{a,c} : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x^2 + c - ay \\ x \end{pmatrix} \qquad (x, y, a, c \in \mathbb{C})$$

By complexification, we can detour around bifurcations:

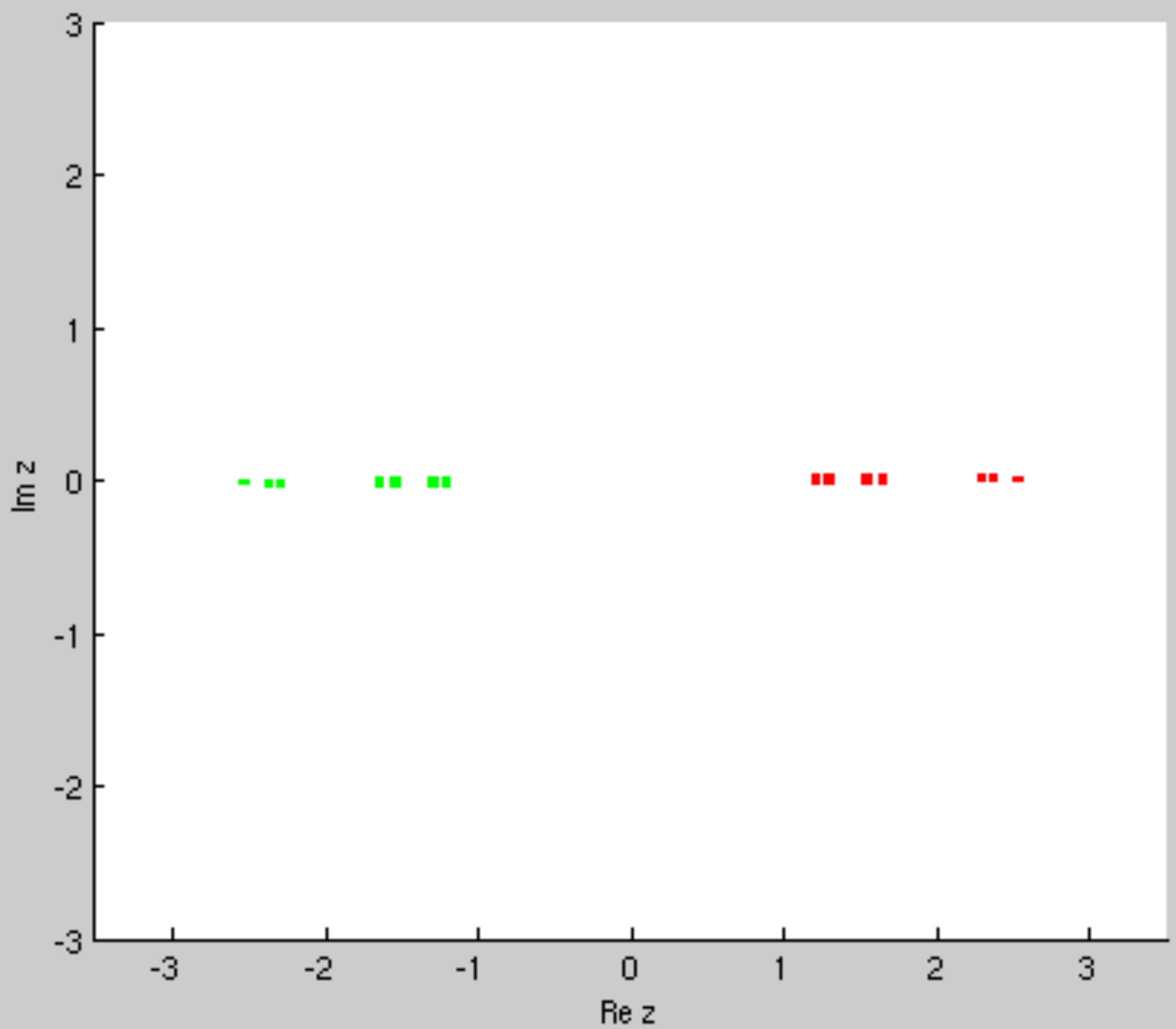


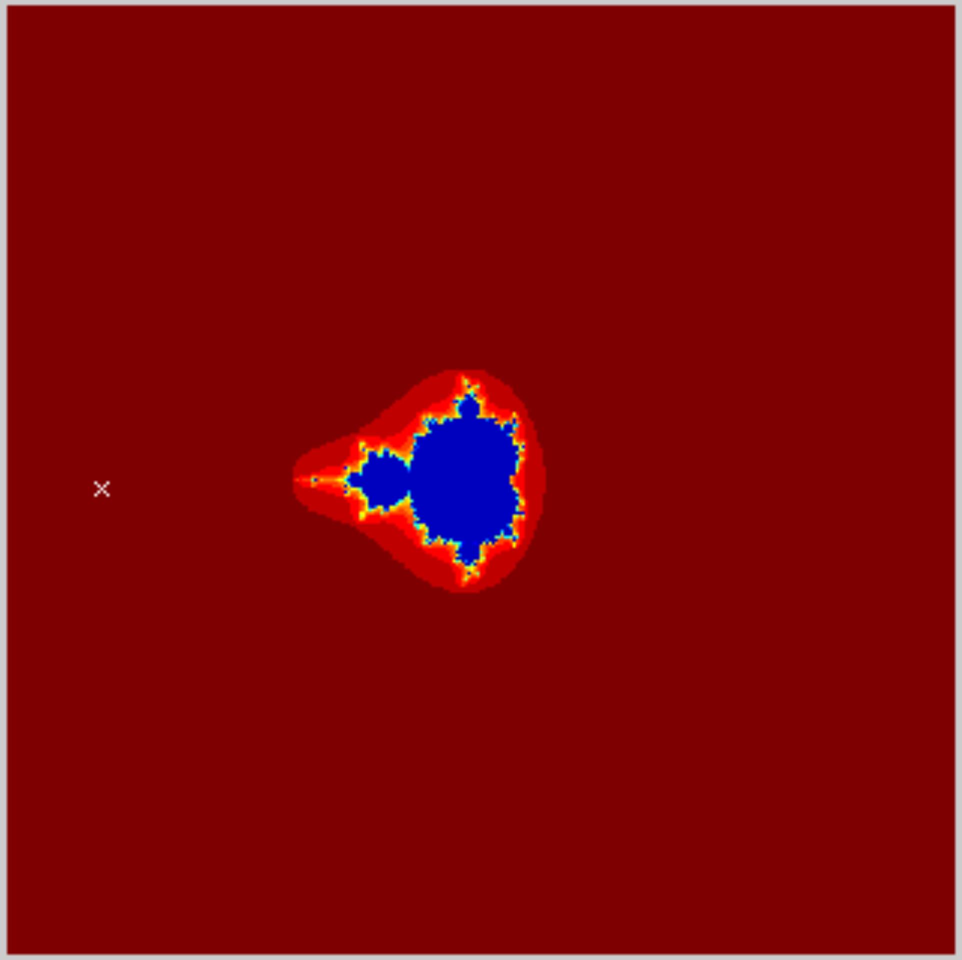
γ : a loop in the complex parameter space

Monodromy

Turning around the M-set

We consider how the filled Julia set K_c moves as the parameter c turns around the Mandelbrot set.





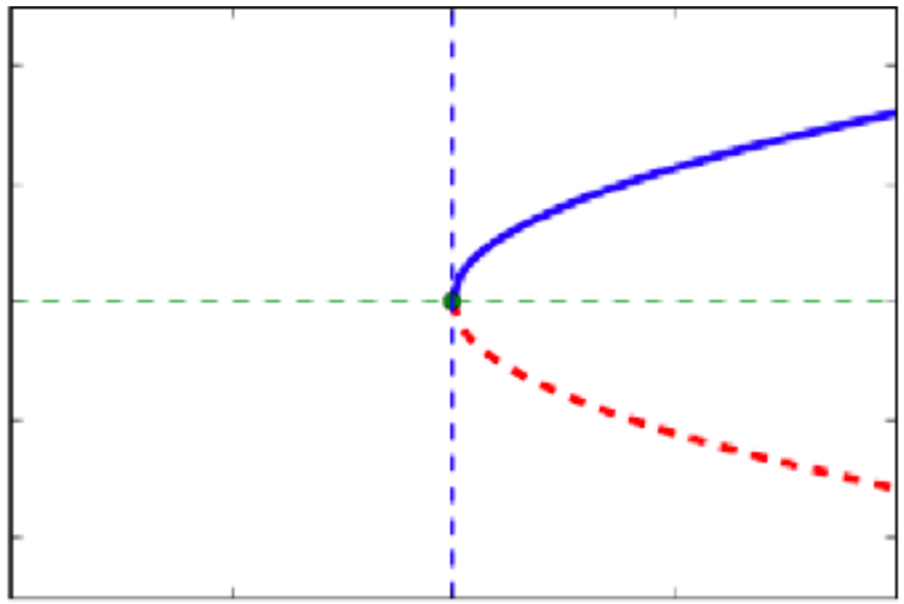
At the base point $c_* = \gamma(0) = \gamma(1)$ of the loop, we have two distinct symbolic codings of K_{c_*}

Monodromy: the easiest case

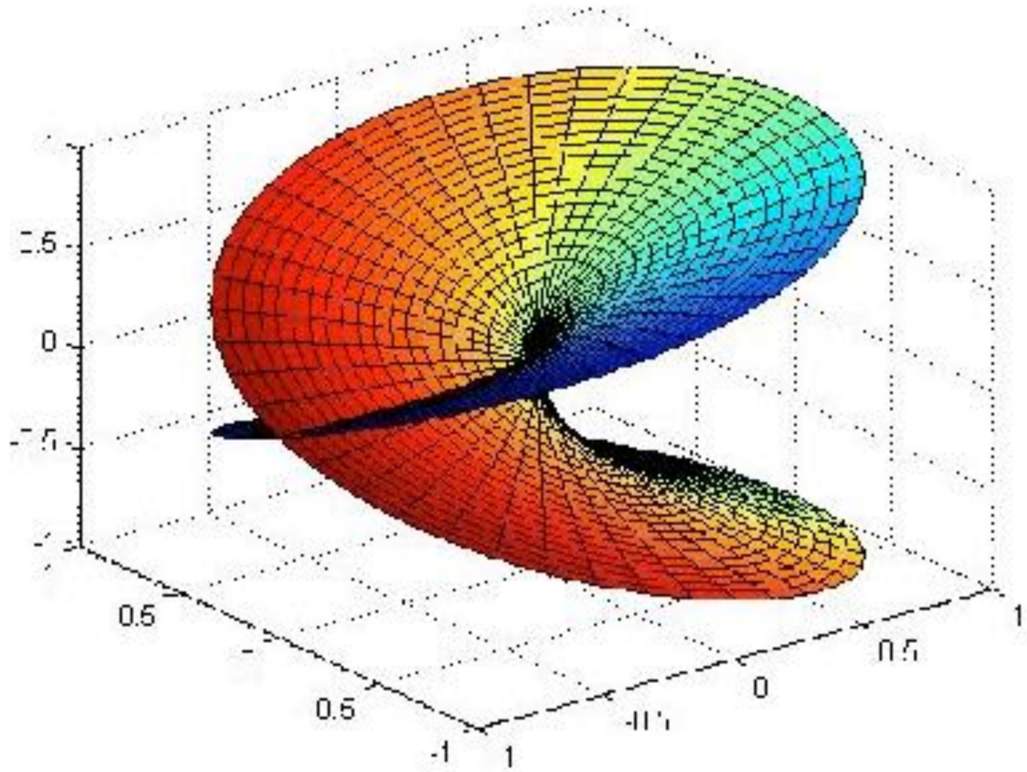
Consider the fixed points of $z \mapsto z^2 + c$.

Sol. of the fixed pt equation: $z = \dfrac{1 \pm \sqrt{1-4c}}{2}$

A saddle node bifurcation takes place at $c = 1/4$.



real saddle-node



complex saddle-node

Monodromy homomorphism 1dim

This defines the monodromy homomorphism

$$\rho : \pi_1(\mathbb{C} \setminus \mathcal{M}) \to \mathrm{Aut}(\Sigma_2^{\mathbb{N}})$$
$$\rho(\gamma) = h_{\gamma(1)} \circ (h_{\gamma(0)})^{-1} : \Sigma_2^{\mathbb{N}} \to \Sigma_2^{\mathbb{N}}$$

A homeomorphism $\psi : \Sigma_2^{\mathbb{N}} \to \Sigma_2^{\mathbb{N}}$ is an automorphism if it commutes with the shift map, i.e. $\sigma \circ \psi = \psi \circ \sigma$.

In this case of the quadratic map,

$$\pi_1(\mathbb{C} \setminus \mathcal{M}) \cong \mathbb{Z} \qquad \mathrm{Aut}(\Sigma_2^{\mathbb{N}}) = \{id, \psi\} \cong \mathbb{Z}_2$$

where ψ is the map that exchanges all 0 and 1.

Monodromy homomorphism is surjective

Monodromy homomorphism 2dim

Similarly we can defines the monodromy homomorphism for the Hénon map

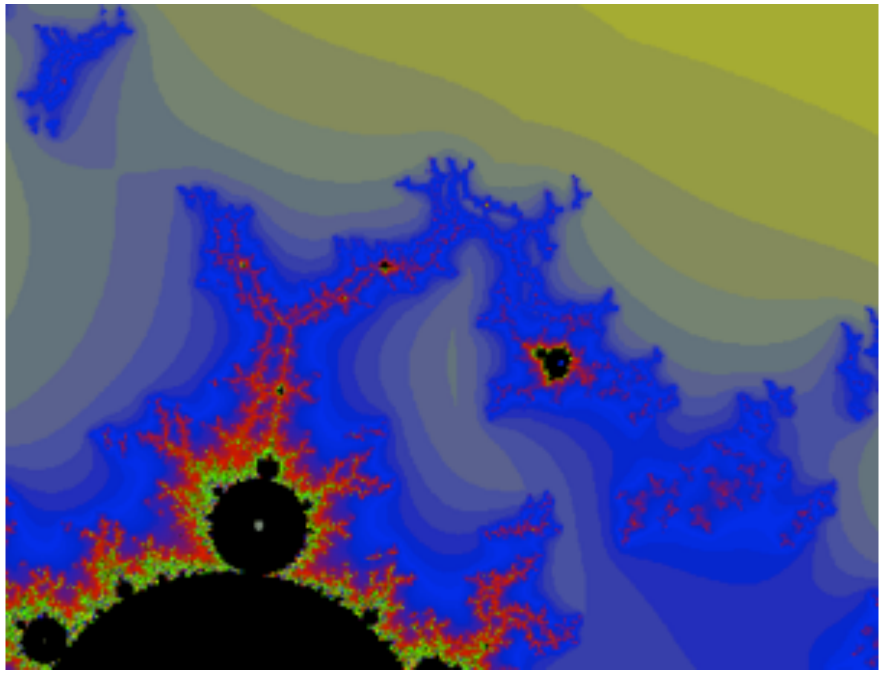
$$\rho : \pi_1(\mathcal{H}) \to \mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$$

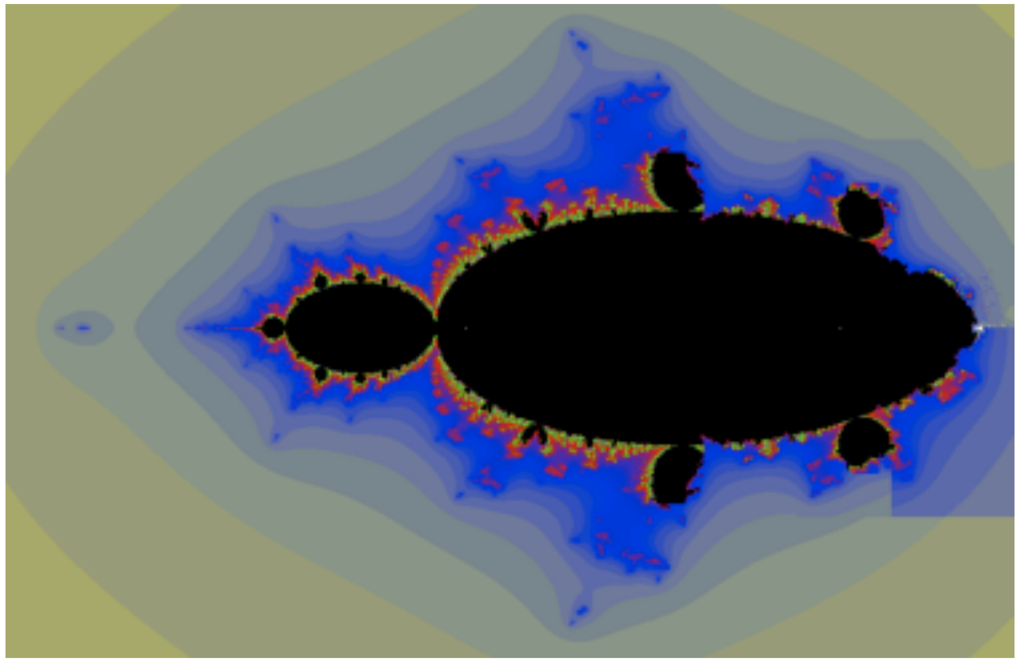
where $\mathcal{H} \subset \mathbb{C}^2$ is the hyperbolic horseshoe parameter locus and $\Sigma_2^{\mathbb{Z}}$ is the two-sided symbolic space.

While $\mathrm{Aut}(\Sigma_2^{\mathbb{N}}) \cong \mathbb{Z}_2$, its two-sided version $\mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$ is extremely complicated. In fact, it contains:

* any finite group
* any countably generated free group
* any finite free product of cyclic group

Hubbard's Conjecture





Numerical Pictures of the param. space of the Hénon map.

Hubbard's Conjecture

These should be "non-trivial loops" in the shift locus. Furthermore, the image of $\rho : \pi_1(\mathcal{H}) \to \mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$ together with the shift map generate $\mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$.

John Hubbard

Results

Let $\mathcal{H}$ be the component of the shift locus of the complex Hénon map containing the real horseshoe region.

Theorem

(1) There are non-trivial loops in $\mathcal{H}$ (and furthermore, the image Γ of the monodromy homomorphism contains an element of infinite order).

(2) (assuming abc conj.) Γ is contained in the set of inert automorphisms. In particular, it does not contain any iteration of σ. This implies that if an automorphism is induced by the monodromy, it satisfies SGCC.

(3) If a loop γ in $\mathcal{H}$ is symmetric w.r.t. complex conjugation ($\bar{\gamma} = \gamma^{-1}$), it induces an involution of $\Sigma_2^{\mathbb{Z}}$.

Rough Sketch of the Proof

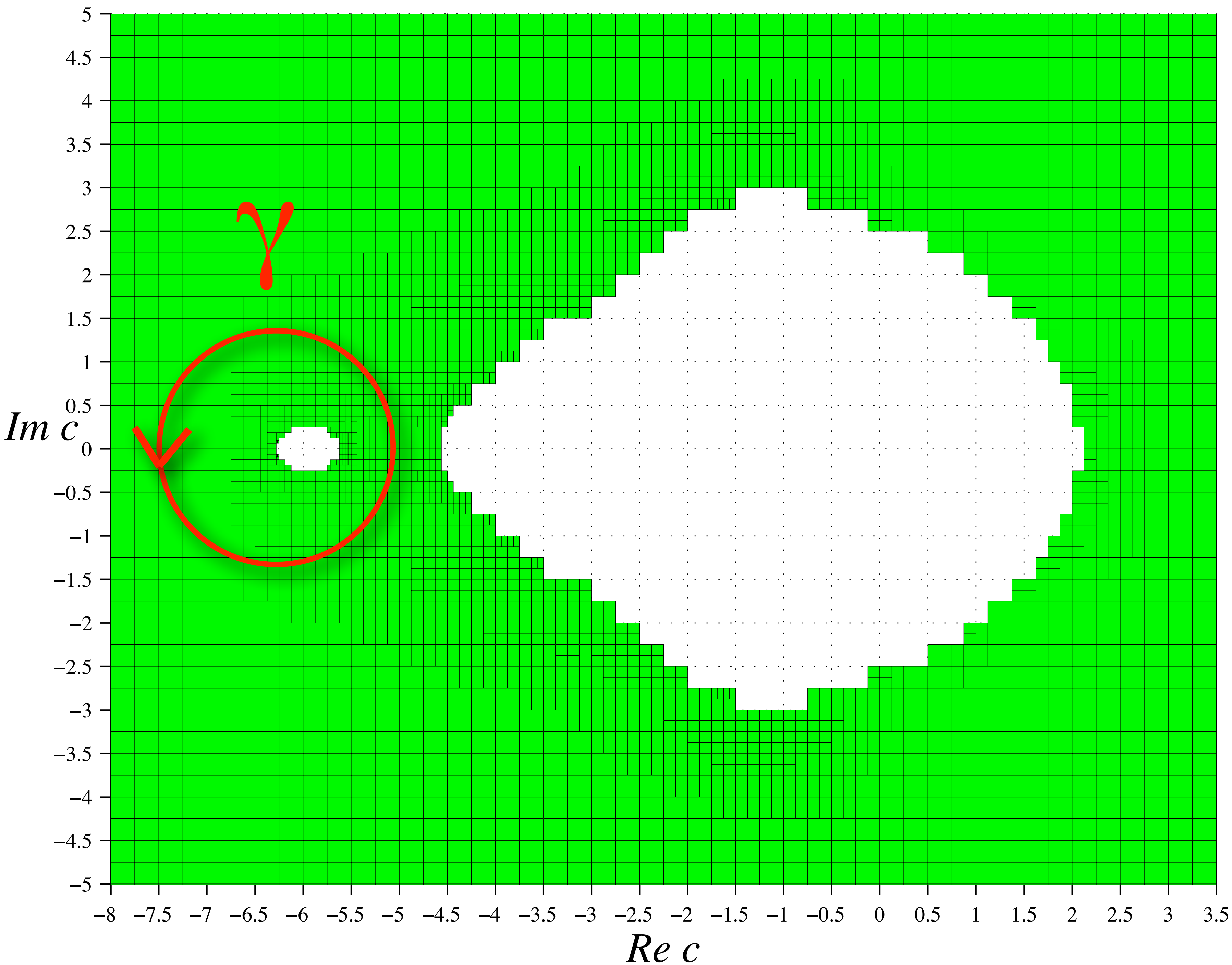
Strategy for (1): Use computer assisted brute-force.

1. Construct a rigorous inner approximation of the shift locus using an algorithm to prove structural stability of the dynamical system.
2. Choose a loop in the computed shift locus and then fix a generating partition at the base point.
3. Compute the monodromy by constructing the continuation of the generating partition

Strategy for (2): Algebraic and Combinatorial

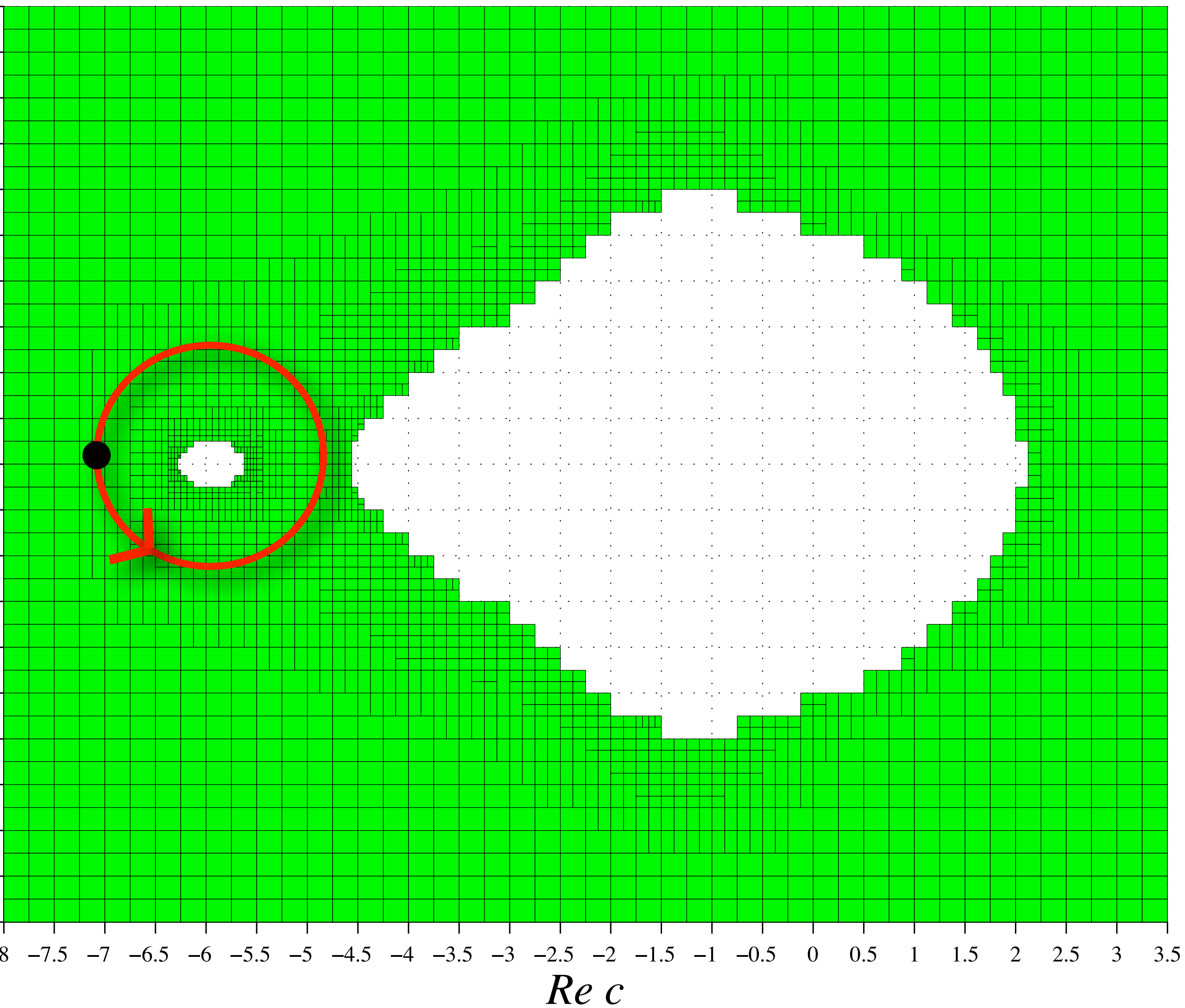
Determine the monodromy action on periodic points by looking at the topology of affine algebraic varieties corresponding to periodic points.

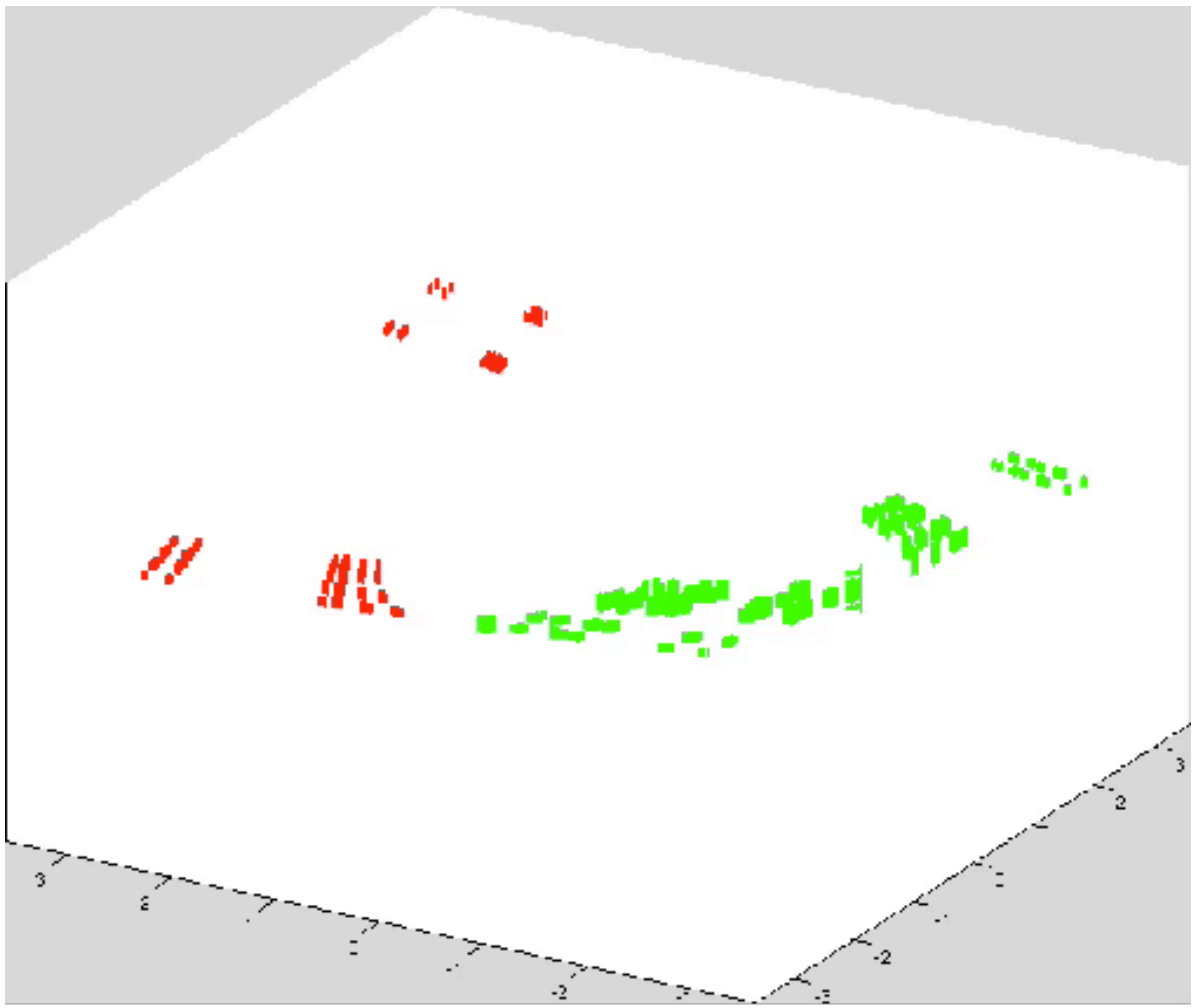
A non-trivial loop



The green region is contained in $\mathcal{H}$ (ZA, Physica D 2016)

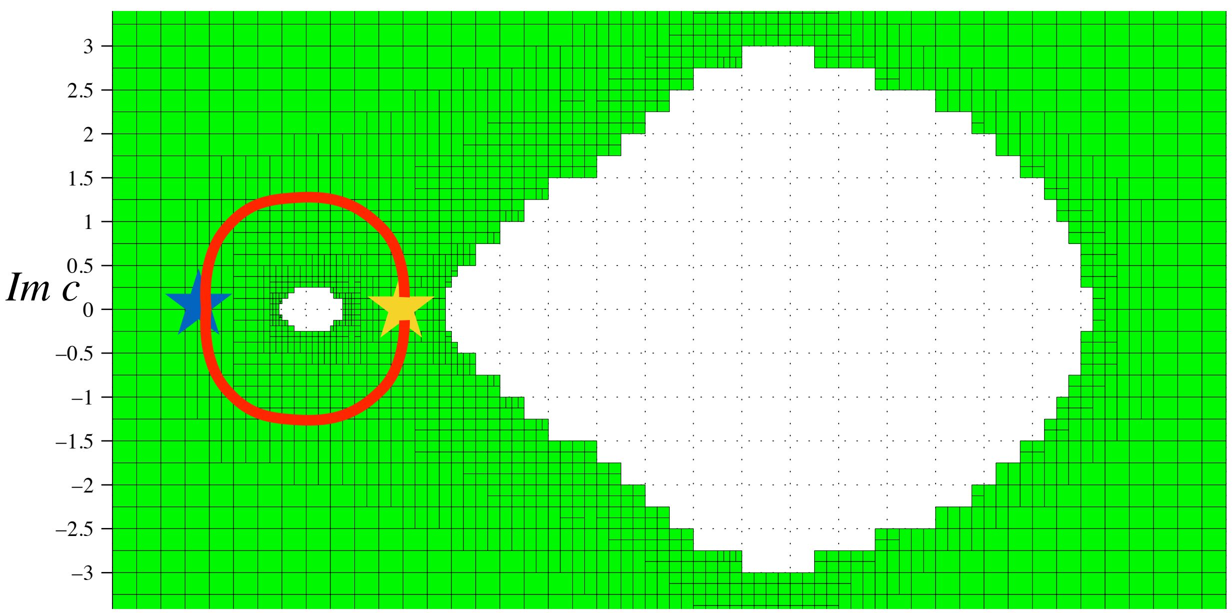
Change of the symbolic coding

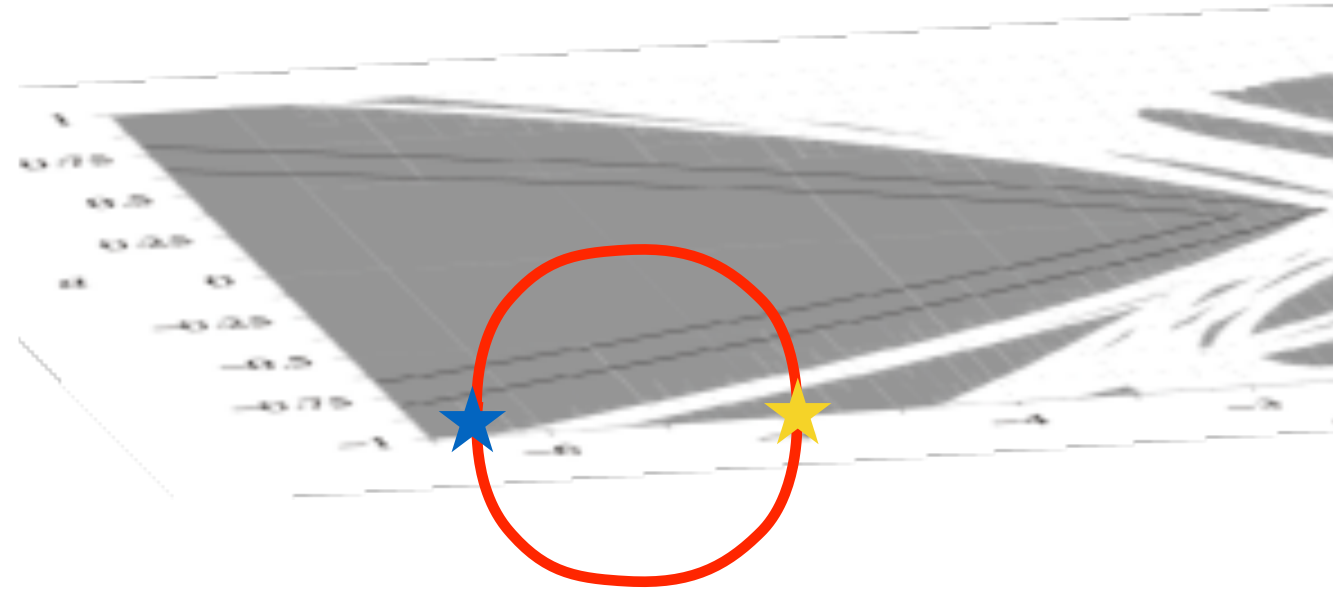




Monodromy Theorem

- ★ (a_0, b_0) : fixed base point in the horseshoe region
- ★ $(a, b) \in \mathcal{H} \cap \mathbb{R}^2$





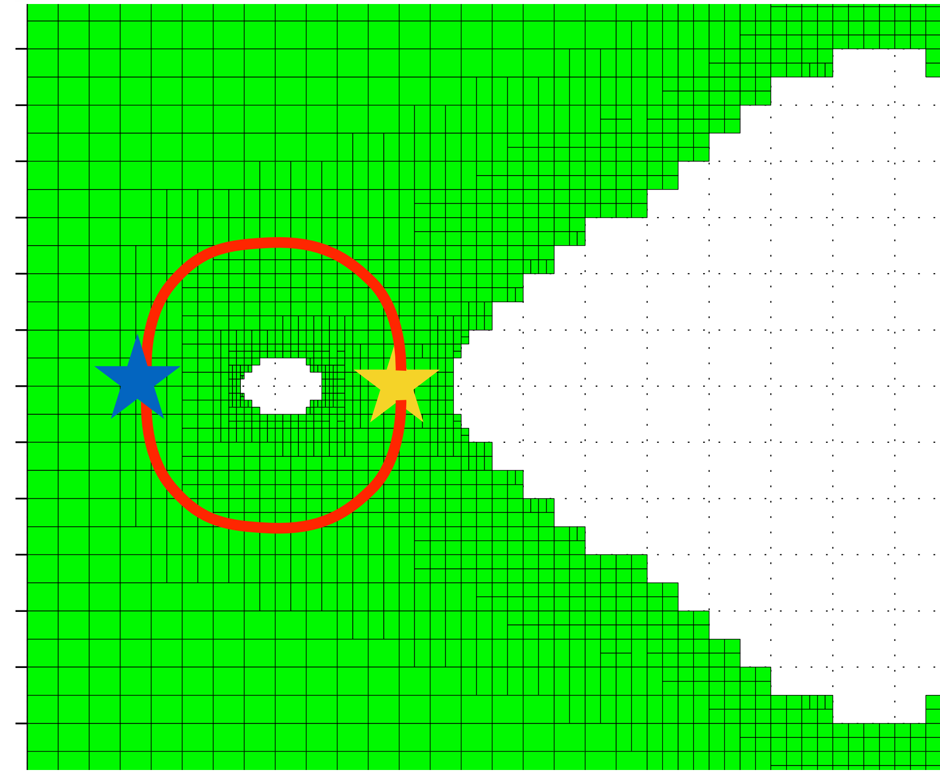
Assume there is a path α in $\mathcal{H}$ connecting these pts.
Define $\gamma = \alpha \cdot (\bar{\alpha})^{-1}$

Then $\rho(\gamma)$ is an involution and $H_{a,c}|_{\mathbb{R}^2} : K^{\mathbb{R}}_{a,c} \to K^{\mathbb{R}}_{a,c}$ is topologically conjugate to

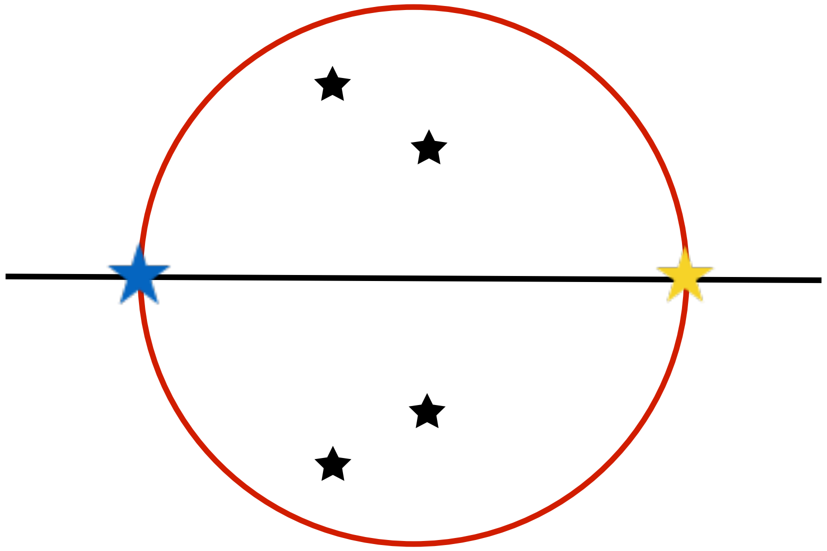
$$\sigma|_{\mathrm{Fix}(\rho(\gamma))} : \mathrm{Fix}(\rho(\gamma)) \to \mathrm{Fix}(\rho(\gamma))$$

Forcing of Real Bifurcation

Thm. For each missing periodic orbit at ★, there should be a **real** bifurcation parameter inside the loop.



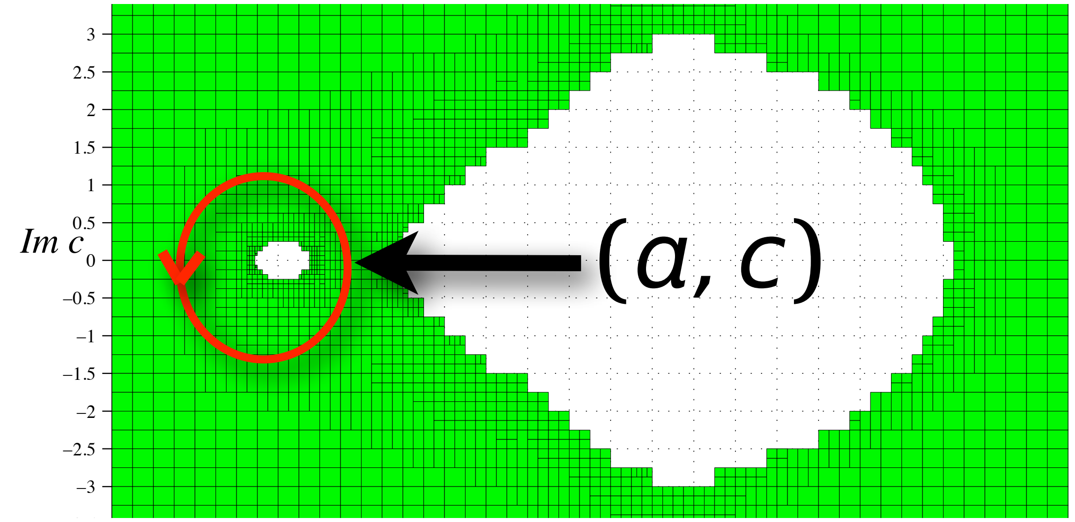
Proof. Otherwise, bifurcations of the orbit take place at pairs of complex conjugate points.



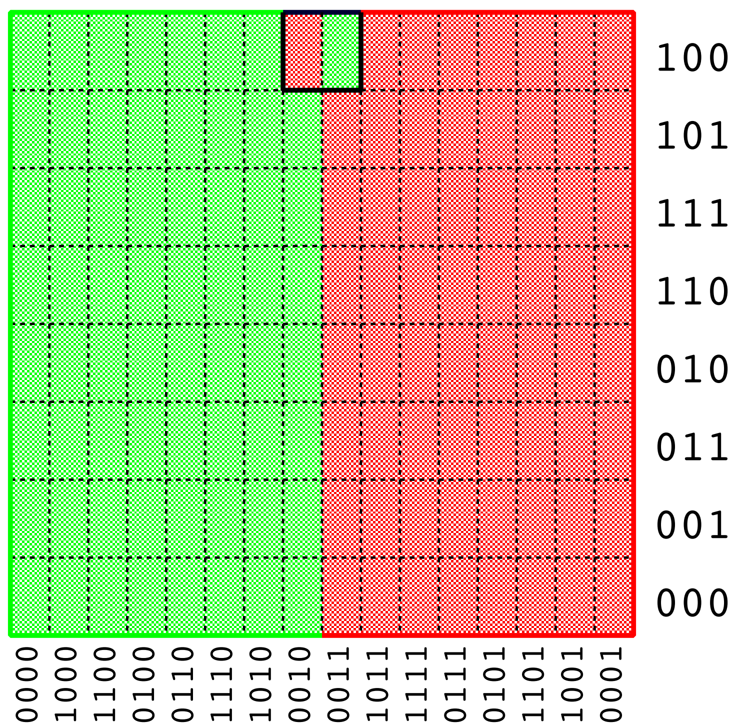
Since the monodromy actions of complex conjugate loops are inverse to each other, the action on the orbit should be trivial. A contradiction.

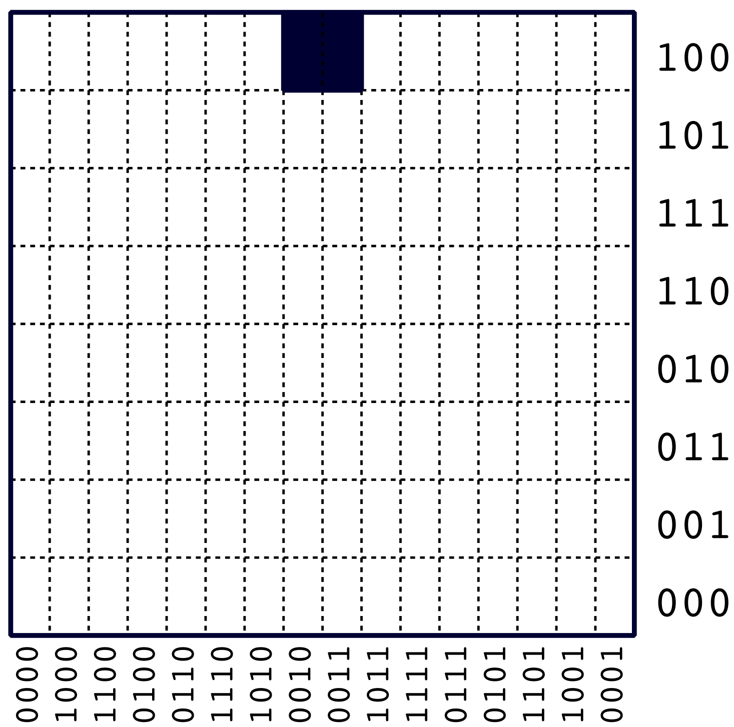
Monodromy and Pruning

The dynamics of the real Hénon map at (a, c) is determined by the following pruning front.



monodromy action





pruning front

That is, the dynamics is the hyperbolic SFT given by two forbidden words 0010100 and 0010111.

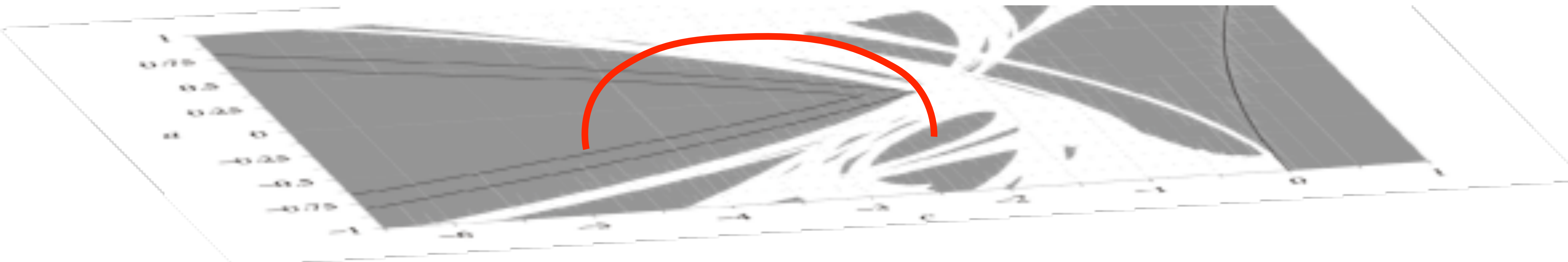
Exact value of the entropy

Corollary. The topological entropy of the real Hénon map on these hyperbolic plateaus are the following:

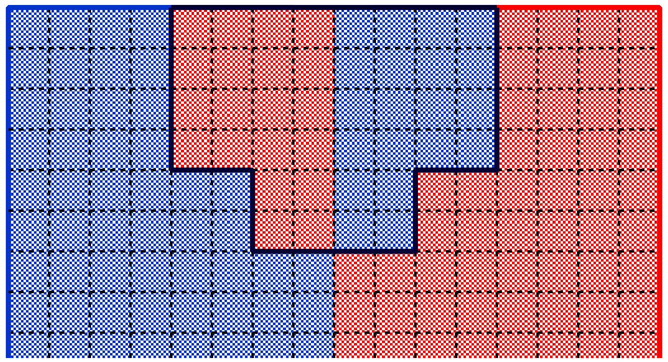
$$h_{\mathrm{top}}(H_{5.4,-1}) = 0.6774440911\ldots$$
$$h_{\mathrm{top}}(H_{2.25,-0.25}) = 0.5967347369\ldots$$
$$h_{\mathrm{top}}(H_{5,1}) = 0.6291888667\ldots$$
$$h_{\mathrm{top}}(H_{2,0.375}) = 0.5403879692\ldots$$

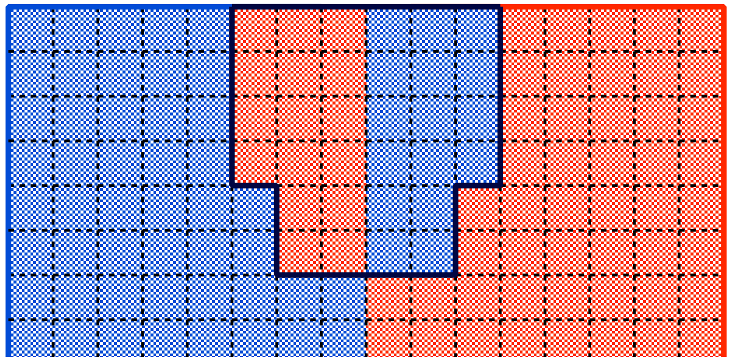
Open Questions

Question 1. Assume that on a real parameter region, the real Hénon is hyperbolic and the complex Hénon is full horseshoe. Is this region connected to the DN region by a path in the complex parameter space?



Question 2. Are pruning front defined by monodromy symmetric and monotonic?





Sign-Gyration Compatibility

SGCC Theorem

Let Γ be the image of $\rho : \pi_1(\mathcal{H}) \to \mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$.

Theorem. If $\phi \in \Gamma$ then ϕ must satisfy the sign-gyration compatibility condition (SGCC) provided that there exist infinitely many non-Wieferich prime numbers.

Remark. For the assumption of the theorem to hold, it suffices to assume the abc conjecture.

The sign number

The n-th sign number $s_n(\phi) \in \mathbb{Z}_2$ of an automorphism $\phi \in \mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$ is

0 iff the permutation induced by ϕ on the set of periodic orbits of least period n is an even permutation.

1 iff the permutation induced by ϕ on the set of periodic orbits of least period n is an odd permutation.

The gyration number

Let $\phi \in \mathrm{Aut}(\Sigma_2^{\mathbb{Z}})$ be an automorphism.

For each periodic orbit U of least period n, we choose an arbitrary element $x_U \in U$.

Since $\phi(x_U)$ and $x_{\phi(U)}$ are in the same periodic orbit, we can find an integer $k(U)$ such that $\phi(x_U) = \sigma^{k(U)}(x_{\phi(U)})$.

Then we define the n-th gyration number $g_n(\phi) \in \mathbb{Z}_n$ by

$$g_n(\phi) := \sum_U k(U) \mod n$$

where the sum is taken over all periodic orbits of least period n.

Sign-Gyration Compatibility

The sign-gyration homomorphism is defined by

$$SGCC_k : \mathrm{Aut}(\Sigma_2^{\mathbb{Z}}) \to \mathbb{Z}_k : \phi \to g_k(\phi) + \sum_{i>0} s_{k/2^i}(\phi)$$

for $k \geq 2$. Here we take $s_{k/2^i} = 0$ whenever $k/2^i$ is not an integer, and for even k we consider $\mathbb{Z}_2$ as subgroup $\{0, k/2\}$ of $\mathbb{Z}_k$.

We say ϕ satisfies the sign-gyration compatibility condition if $SGCC_k(\phi) = 0$ for all $k \geq 2$.

Facts on SGCC auto

For example, $SGCC_2(\phi) = 0$ implies

$$g_2(\phi) = \begin{cases} 0 & \text{iff} \quad s_1(\phi) = 0 \\ 1 & \text{iff} \quad s_1(\phi) = 1 \end{cases}.$$

This implies that if $\phi : \Sigma_2 \to \Sigma_2$ satisfies SGCC, then it rotates the periodic orbit $\overline{01} \in \Sigma_2$ of period 2 if and only if it interchanges two fixed points $\overline{0}$ and $\overline{1}$.

* SGCC holds for any finite composition of elements of finite order.
* The shift map does NOT satisfy SGCC.
* SGCC holds for all inert automorphisms.

FACT: $\mathrm{Aut}(\Sigma_2^{\mathbb{Z}}) \cong \mathbb{Z}\langle\sigma\rangle \oplus \mathrm{Inert}(\sigma_2^{\mathbb{Z}})$

Periodic Point Surfaces

For $n \geq 1$, define polynomials F_n and G_n by $H^n_{a,c}(x,y) = (F_n(x,y,a,c), G_n(x,y,a,c))$.

For a periodic sequence $s \in \Sigma_2$ of length n, denote by $\tilde{X}_s$ the component of the algebraic variety ($\subset \mathbb{C}^4$) defined by

$$F_n(x,y,a,c) - x = 0, \quad G_n(x,y,a,c) - y = 0$$

containing $h_{(a_0,c_0)}(s)$ and by X_s the intersection of $\tilde{X}_s$ and the algebraic variety defined by

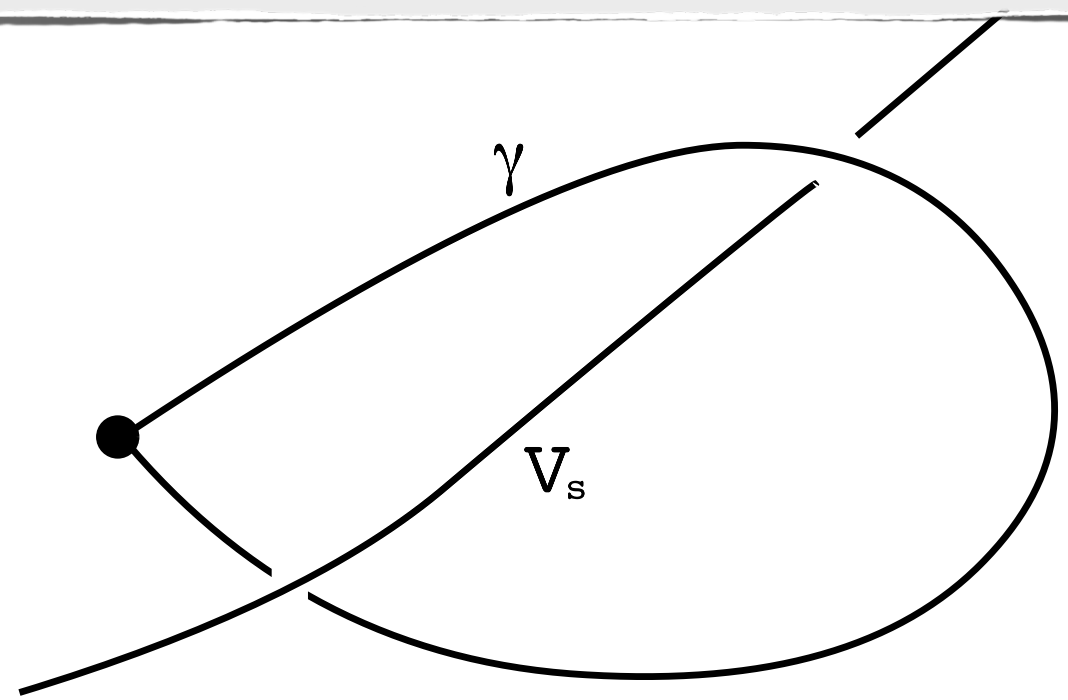
$$F_n(x,y,a,c) - x = 0$$
$$G_n(x,y,a,c) - y = 0$$
$$\det \begin{pmatrix} \frac{dF_n}{dx} - 1 & \frac{dF_n}{dy} \\ \frac{dG_n}{dx} & \frac{dG_n}{dy} - 1 \end{pmatrix} = 0$$

Monodromy and Bifurcation

Let V_s be the projection of X_s to (a, c) space.
We can show that V_s is an affine algebraic variety.

Proposition

Let $s \in \Sigma_2$ and γ a loop in $\mathcal{H}$. Then $\rho(\gamma)(s)$ depends only on the homotopy class $[\gamma] \in \pi_1(\mathbb{C}^2 \setminus V_s)$.
In particular, if $[\gamma]$ is trivial in $\pi_1(\mathbb{C}^2 \setminus V_s)$, then s is fixed by $\rho(\gamma)$.



Lemma

Let $\phi \in \Gamma$. Then $\mathrm{SGCC}_2(\phi) = 0$.

Proof. Let $\phi = \rho(\gamma)$. We can show that

$$V_{\overline{0}} = V_{\overline{1}} = \{(a,c) \mid c = \frac{1}{4}(b+1)^2\} \text{ and}$$

$$V_{\overline{01}} = \{(a,c) \mid c = -\frac{3}{4}(b+1)^2\}$$

are homotopic in $\mathbb{C}^2 \setminus \mathcal{H}$ via $V_\lambda = \{c = \lambda(b+1)^2\}$ and therefore γ determines the same element in $\pi_1(\mathbb{C}^2 \setminus V_{\overline{0}})$ and $\pi_1(\mathbb{C}^2 \setminus V_{\overline{01}})$. By a direct computation, it follows that ϕ interchanges $\overline{0}$ and $\overline{1}$ $\Leftrightarrow$ $\gamma \in \pi_1(\mathbb{C}^2 \setminus V_{\overline{0}})$ is odd $\Leftrightarrow$ ϕ rotates $\overline{01}$.

Lemma

Let $\phi \in \Gamma$. Then $SGCC_p(\phi) = 0$ for all odd prime p.

It suffices to show $g_p(\phi) = 0$. Let $\phi = \rho(\gamma)$.

Let M_1 be the set of parameters for which $\exists$ fixed point s.t. both of its eigen values are of modulus ≥ 1 or ≤ 1 .

Step 1. Only X_s contained in M_1 contributes to $g_p(\phi)$.

Step 2. For any $\gamma \in \pi_1(\mathbb{C} \setminus M_1)$, show γ^2 is homotopic to a standard 2k-fold loop around M_1 contained in a slice $\{a = \text{real const.}\}$

Step 3. Show $g_p(\rho(\gamma^2)) = 0$ mod p $\Rightarrow$ $g_p(\rho(\gamma)) = 0$

Dim. Group Representation

For a subshift Σ_A of finite type defined by a matrix A, the inert automorphism subgroup $\mathrm{Inert}(\Sigma_A)$ is defined to be the kernel of "dimension group representation"

$$\delta_A : \mathrm{Aut}(\Sigma_A) \to \mathrm{Aut}(s_A).$$

When Σ_A is a full shift, δ_A is surjective and the shift is mapped to the generator of $\mathrm{Aut}(s_A) \cong \mathbb{Z}$.

In this case,

$$1 \to \mathrm{Inert}(\Sigma_A) \to \mathrm{Aut}(\Sigma_A) \to \mathbb{Z} \to 1$$

splits and since the shift commutes with all other automorphisms, we write

$$\mathrm{Aut}(\Sigma_A) = \mathbb{Z}\langle\sigma\rangle \oplus \mathrm{Inert}(\Sigma_A).$$

Wieferich Prime Number

Wieferich prime

Wieferich prime is a prime number p such that p^2 divides $2^{p-1} - 1$.

- Fermat's little thm says that p divides $2^{p-1} - 1$.
- The only known Wieferich primes are: 1093, 3511.
- There is no other Wieferich prime < 6.7×10^{15}.

It is not known if there exists infinitely many Wieferich number or non-Wieferich number.

abc Conjecture

Silverman's Theorem

If the abc conjecture is true, then there exists infinitely many non-Wieferich prime numbers.

abc conjecture

For any $\epsilon > 0$, there exist only finitely many triples of positive coprime integers (a, b, c) such that

$$a + b = c \quad \text{and} \quad c > d^{1+\epsilon}$$

where d is the radical of abc.

Wieferich prime and SGCC

Lemma: For an odd prime p, $\mathrm{SGCC}_p(\sigma) = 0$ if and only if p is a Wieferich prime.

Proof. The number of p-periodic orbits is

$$(2^p - 2)/p = 2(2^{p-1} - 1)/p$$

Since each periodic orbit is gyrated one-step ahead by the shift map, the gyration number is

$$2(2^{p-1} - 1)/p \mod p,$$

which is 0 exactly when p is Wieferich.

Proof of Theorem

Let $\phi \in \Gamma$ and assume $\phi = (\sigma^k, \phi') \in \mathbb{Z} \oplus \mathrm{Inert}(\Sigma_2)$.
Assume $k \neq 0$.

Choose a non-Wieferich prime p which is coprime to k.
Since p is a prime, $\mathrm{SGCC}_p(\phi) = 0$.

Since p is non-Wieferich and coprime to k, we have $\mathrm{SGCC}_p(\sigma^k) = k \cdot \mathrm{SGCC}_p(\sigma) \neq 0$.

On the other hand, we have $\mathrm{SGCC}_p(\phi') = 0$. Thus

$$\mathrm{SGCC}_p(\phi) = \mathrm{SGCC}_p(\sigma^k) + \mathrm{SGCC}_p(\phi') \neq 0$$

which is a contradiction. Therefore $k = 0$.

Open Questions

Question. Can we prove the theorem directly for all period p, without assuming abc conjecture?

Question. Is there any relation between the algebraic restriction of SGCC and the geometric description of monodromies by Christopher Lipa?

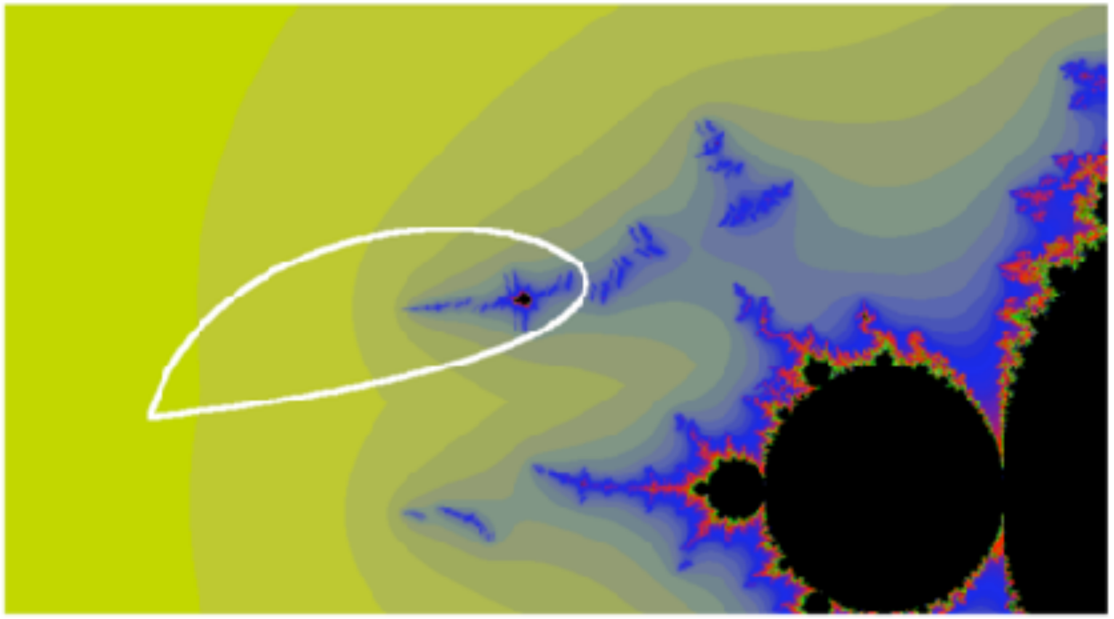


Figure 9.10: Loop around A herd of $\mathcal{W}_{\mathrm{EAB3}}$

Connectedness
and
Disconnectedness

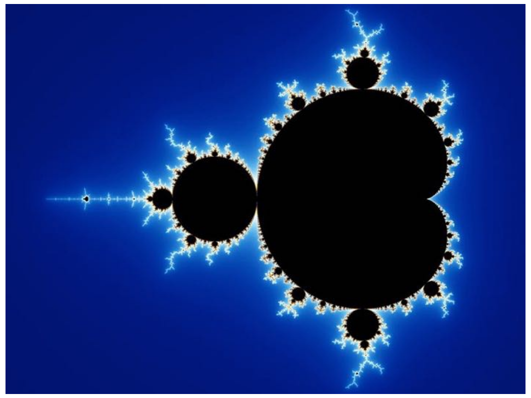
Mandelbrot Set

The Mandelbrot Set $\mathscr{M} := \{c \in \mathbb{C} \mid \{P_c^n(0)\} \text{ is bounded}\}$

Dichotomy:

A. If $c \in \mathscr{M}$ then J_c and K_c are connected

B. If $c \notin \mathscr{M}$ then $J_c = K_c$ and is homeo to $\Sigma_2^{\mathbb{N}}$



FACT: $\mathscr{M}$ is connected

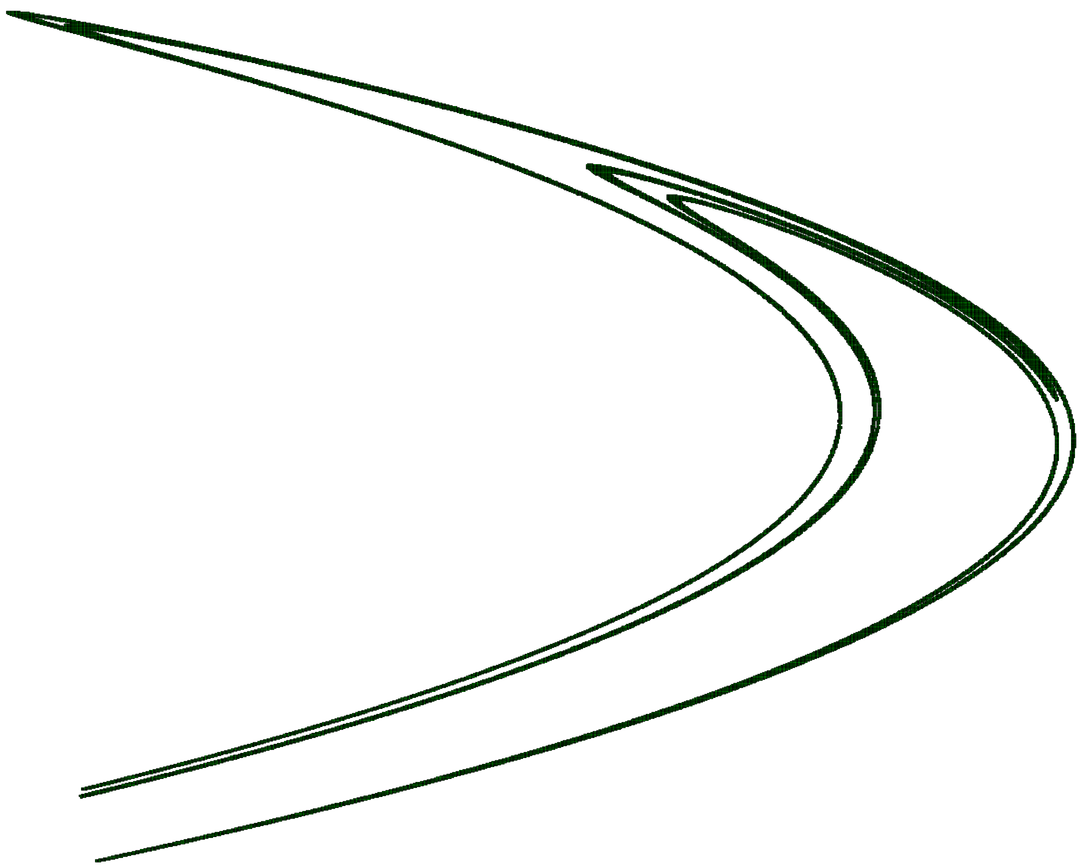
2dim case (Hénon map): $f_{a,c}(x, y) = (x^2 + c - ay, x)$

Let $\mathscr{M} := \{(a, c) \mid J_{a,c} \text{ is connected}\}$

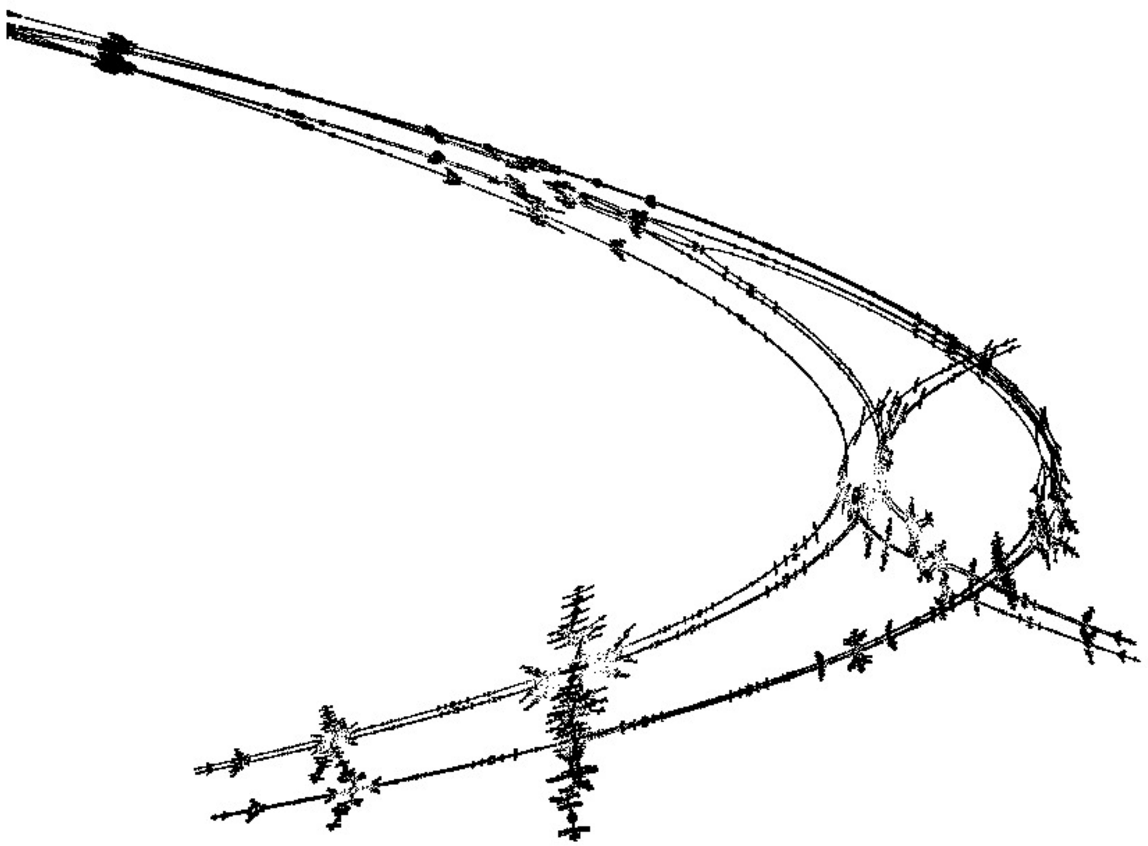
Q: Is $\mathscr{M}$ connected?

REMARK: Since there is no critical point, we need to discuss the connectedness of J_c another way.

Julia set for the Hénon map

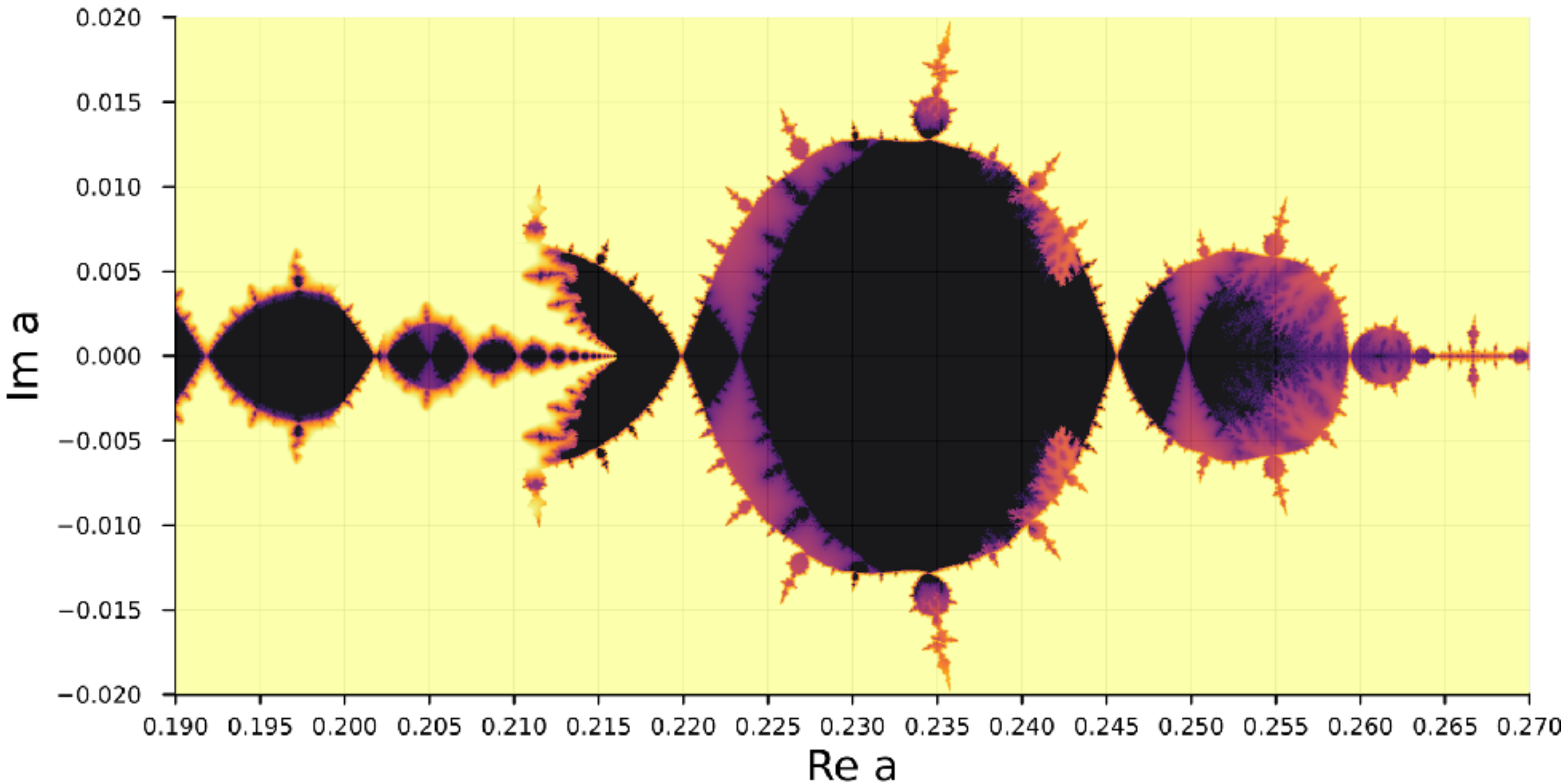


the strange attractor
(real Hénon map)



its complexification
(by S. Ushiki)

Complex Parameter Space



Non-rigorous picture of parameter space with $c = -1.29$

Black: connectedness locus

Yellow: full-shift locus

Periodic Points

Periodic Points

x is a periodic point of period n if $f^n(x) = x$

Hyperbolic Periodic Points

x : a periodic point of period n

x is transversal $\Leftrightarrow$ $Df^n(x)$ doesn't have eigenvalue 1.

x is hyperbolic $\Leftrightarrow$ $Df^n(x)$ doesn't have eigenvalue of absolute value 1.

Stability of Hyperbolic Periodic Points

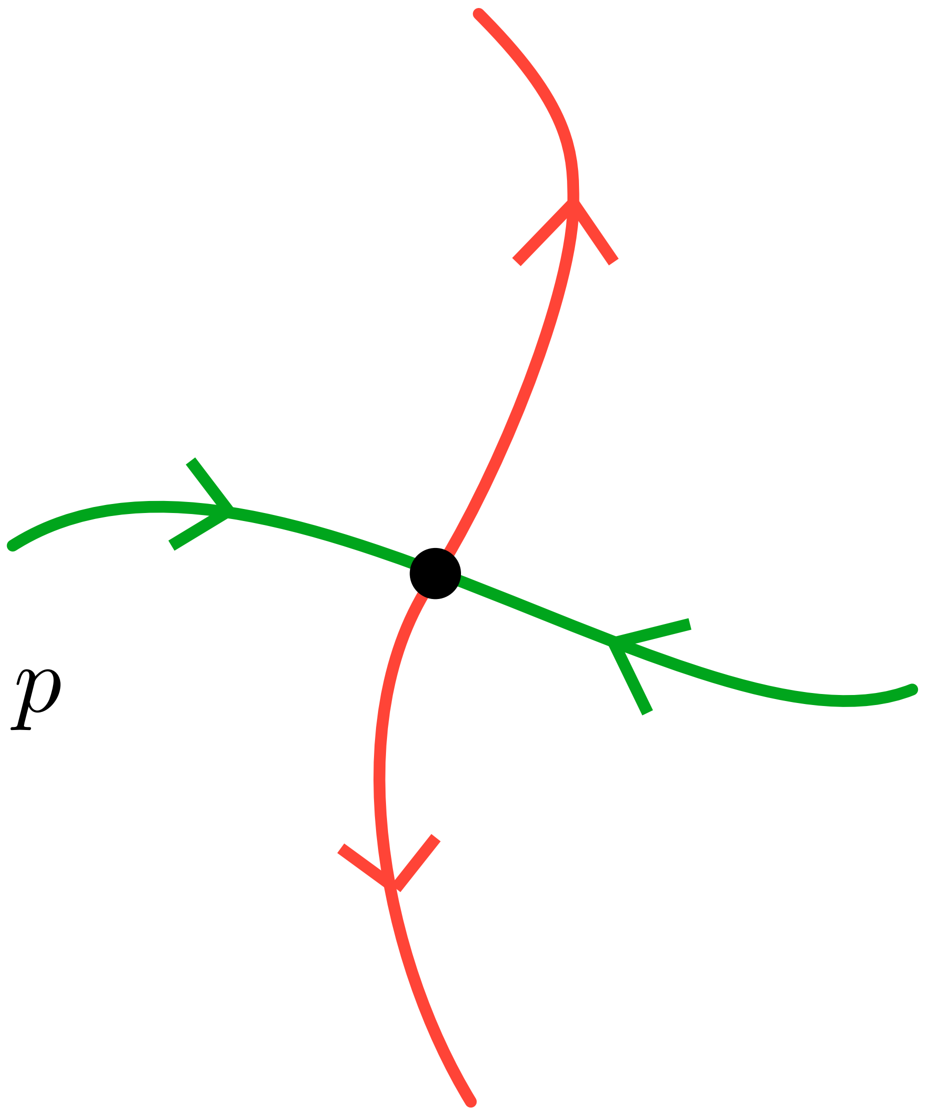
transversal p.p. $\Rightarrow$ p.p is robust w.r.t. small perturbation

hyperbolic p.p. $\Rightarrow$ the dynamics on a nbd of the periodic point is robust w.r.t small perturbation

(un)stable manifolds

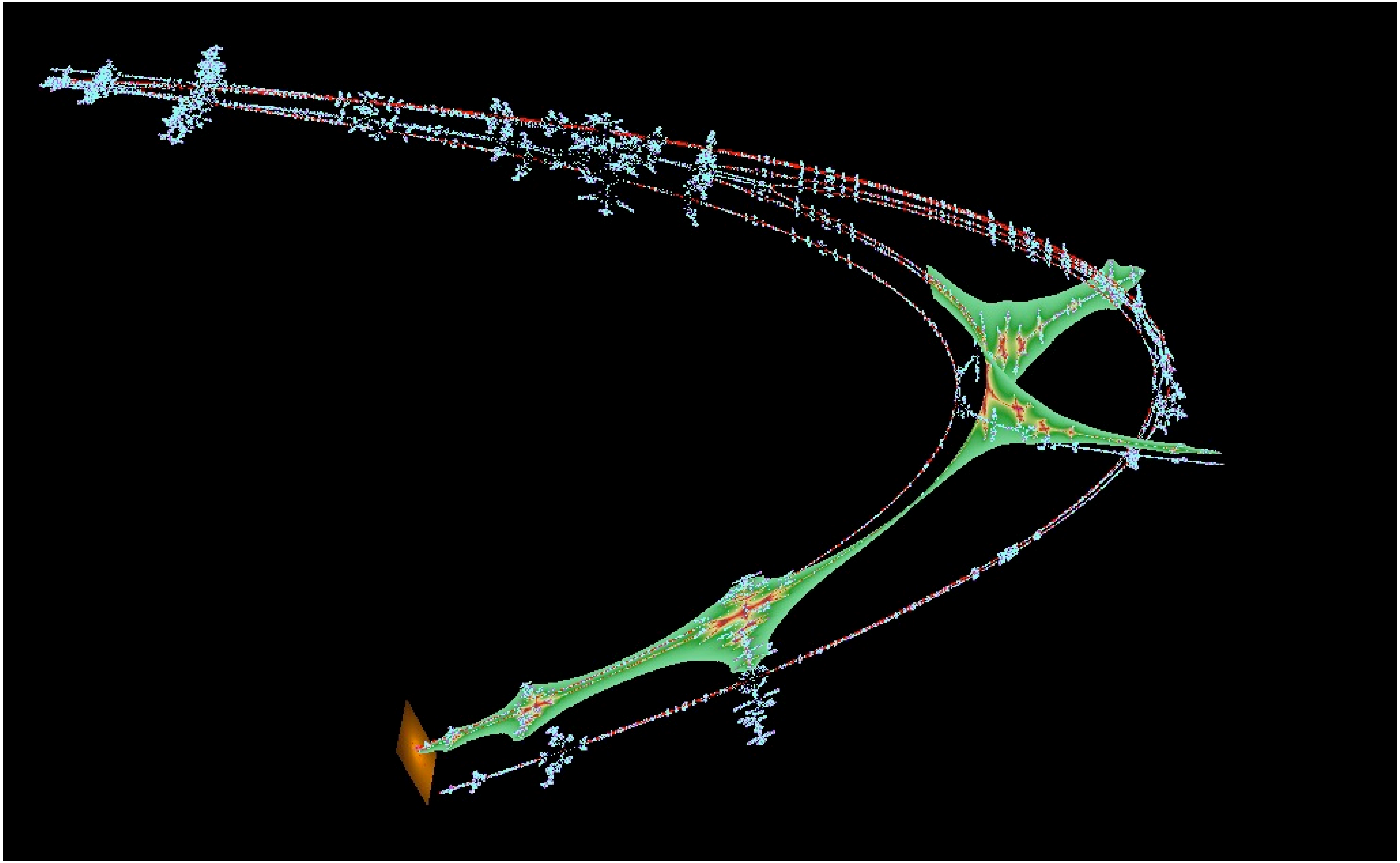
For a hyperbolic periodic point p, define

$$W^s(p) := \{x \in M \mid d(f^n(x), p) \to 0 \text{ as } n \to \infty\}$$
$$W^u(p) := \{x \in M \mid d(f^n(x), p) \to 0 \text{ as } n \to -\infty\}$$



They are smooth (immersed) submanifolds of the phase space called stable/unstable manifold of p

Complex (un)stable manifold

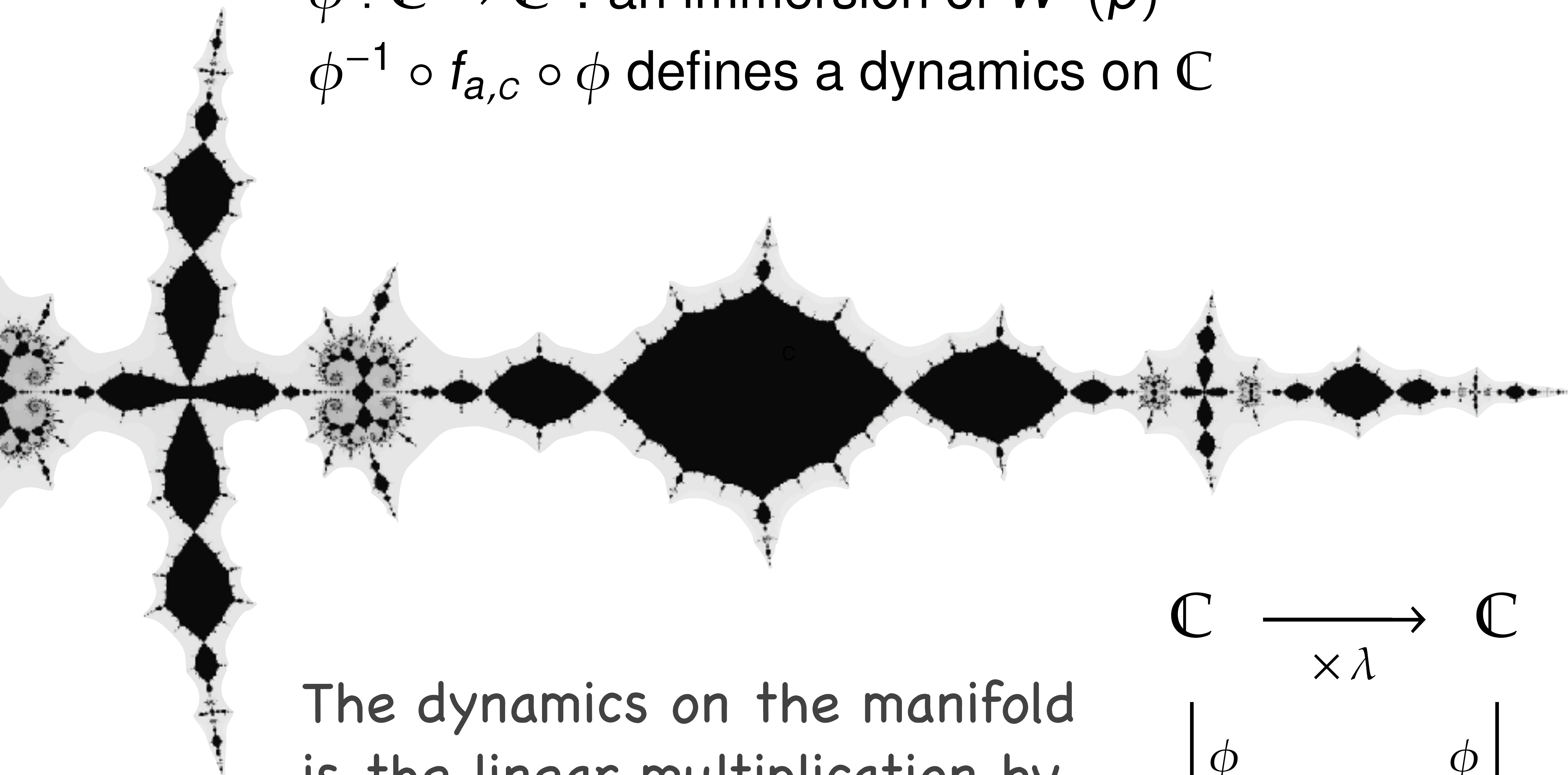


Complex (un)stable mfds by Shigehiro Ushiki

Julia Slice on the Unstable Manifold

$\phi : \mathbb{C} \to \mathbb{C}^2$: an immersion of $W^u(p)$

$\phi^{-1} \circ f_{a,c} \circ \phi$ defines a dynamics on $\mathbb{C}$



The dynamics on the manifold is the linear multiplication by the unstable eigenvalue λ

$$\begin{tikzcd} \mathbb{C} \arrow[r, "\times\lambda"'] \arrow[d, "\phi"] & \mathbb{C} \arrow[d, "\phi"'] \\ \mathbb{C}^2 \arrow[r, "f_{a,c}"] & \mathbb{C}^2 \end{tikzcd}$$

The Green function and the maximum principle

Green Function

We define the Green function for $f = f_{a,c}$ by

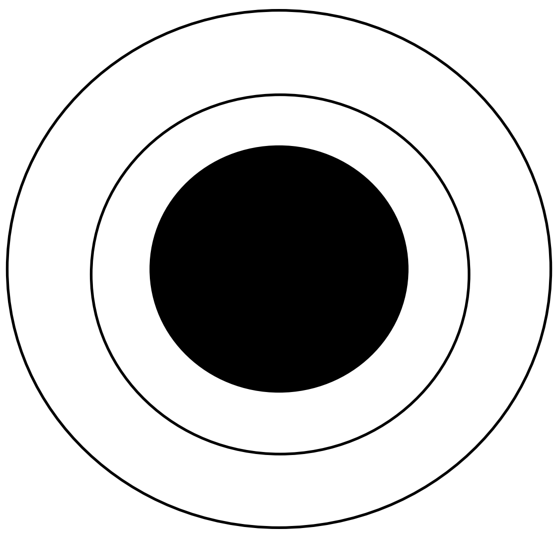
$$G^+(x,y) = \lim_{n\to\infty} \frac{1}{2^n} \log^+ \|f^n(x,y)\|$$

where $\log^+ z = \max\{\log z, 0\}$.

- G^+ is (pluri)harmonic on $\mathbb{C}^2 \setminus K^+$.
- $G^+ = 0$ on K^+.
- G^+ is (pluri)subharmonic on the entire $\mathbb{C}^2$.

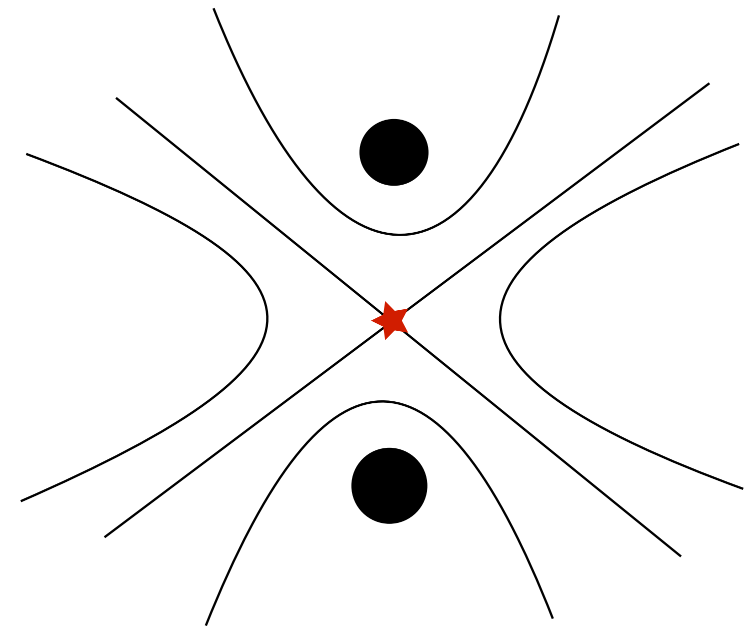
Critical Point of Green Function

connected



No critical point

disconnected



Critical point appears

Proposition (1-dim case)

Consider $P_c(z) = z^2 + c$. If z is a critical point of G^+ which is not contained in the filled Julia set, then $\exists m \geq 0$ such that $P_c^m(z) = 0$. Therefore,

$$\exists \text{ a crit. point in } \mathbb{C} \setminus K_c \quad \Rightarrow \; 0 \notin K_c \; \Rightarrow \; K_c \text{ is disconnected.}$$

A Criterion for 2D

Theorem (Bedford-Smillie, Dujardin)

Let p be a saddle point of f. Then the followings are equivalent.

- *J_f is connected*
- *the restriction of G^+ on $W^u(p) \setminus K_f$ has no critical point.*
- *$W^u(p) \cap K_f$ has no compact component*

Let $\phi : \mathbb{C} \to \mathbb{C}^2$ the analytic function for $W^u(p)$, that is,

$$\phi(0) = p, \quad \phi'(0) = \lambda, \quad f(\phi(z)) = \phi(\lambda(z))$$

where λ is the unstable eigen value at p.

To prove the disconnectedness of J_f, it suffices to show the existence of a critical point of $G^+ \circ \phi$.

A numerical search for the critical point is relatively easy, but...

Bound for the unstable manifold

Let $\phi(z) = (x(z), y(z))$ and

$$x(z) = s_0 + s_1 z + s_2 z^2 + s_3 z^3 + \cdots$$
$$y(z) = t_0 \ + t_1 z \ + t_2 z^2 \ + t_3 z^3 + \cdots$$

be the Taylor expansion of x and y.

Then (s_0, t_0) and (s_1, t_1) is determined by p and the eigenvector.

For $n > 1$, it follows from $f(\phi(z)) = \phi(\lambda(z))$ that

$$s_n = \frac{\sum_{i=1}^{n-1} s_i s_{n-i}}{\lambda^n + \frac{a}{\lambda^n} - 2s_0}$$
$$t_n = s_n / \lambda^n.$$

We also have a rigorous estimate of the reminder term.

Bounds for the Green function

Let

$$R_0 = \frac{1 + |a| + \sqrt{(1 + |a|)^2 + 4|c|}}{2}$$

$$R_1 = \max\{R_0, |a| + \sqrt{|a|^2 + 2|c|}\}$$

and define

$$V_{R_0} = \left\{(x, y) \in \mathbb{C}^2 : |x| \geq R_0,\ |x| \geq |y|\right\}$$

$$V_{R_1} = \left\{(x, y) \in \mathbb{C}^2 : |x| \geq R_1,\ |x| \geq |y|\right\}.$$

Lemma

1. $f^n(x_0, y_0) \in V^+_{R_0} \Rightarrow G^+(x_0, y_0) \leq \frac{1}{2^n}(\log|x_n| + \log 2)$,
2. $f^n(x_0, y_0) \in V^+_{R_1} \Rightarrow G^+(x_0, y_0) \geq \frac{1}{2^n}(\log|x_n| - \log 2)$.

Maximum (Minimum) Principle

Lemma

Let $C > 0$ and γ be a loop in $\mathbb{C}$ such that

$$G^+(\phi(z)) > C$$

for all $z \in \gamma$ and assume there is z_0 inside γ such that

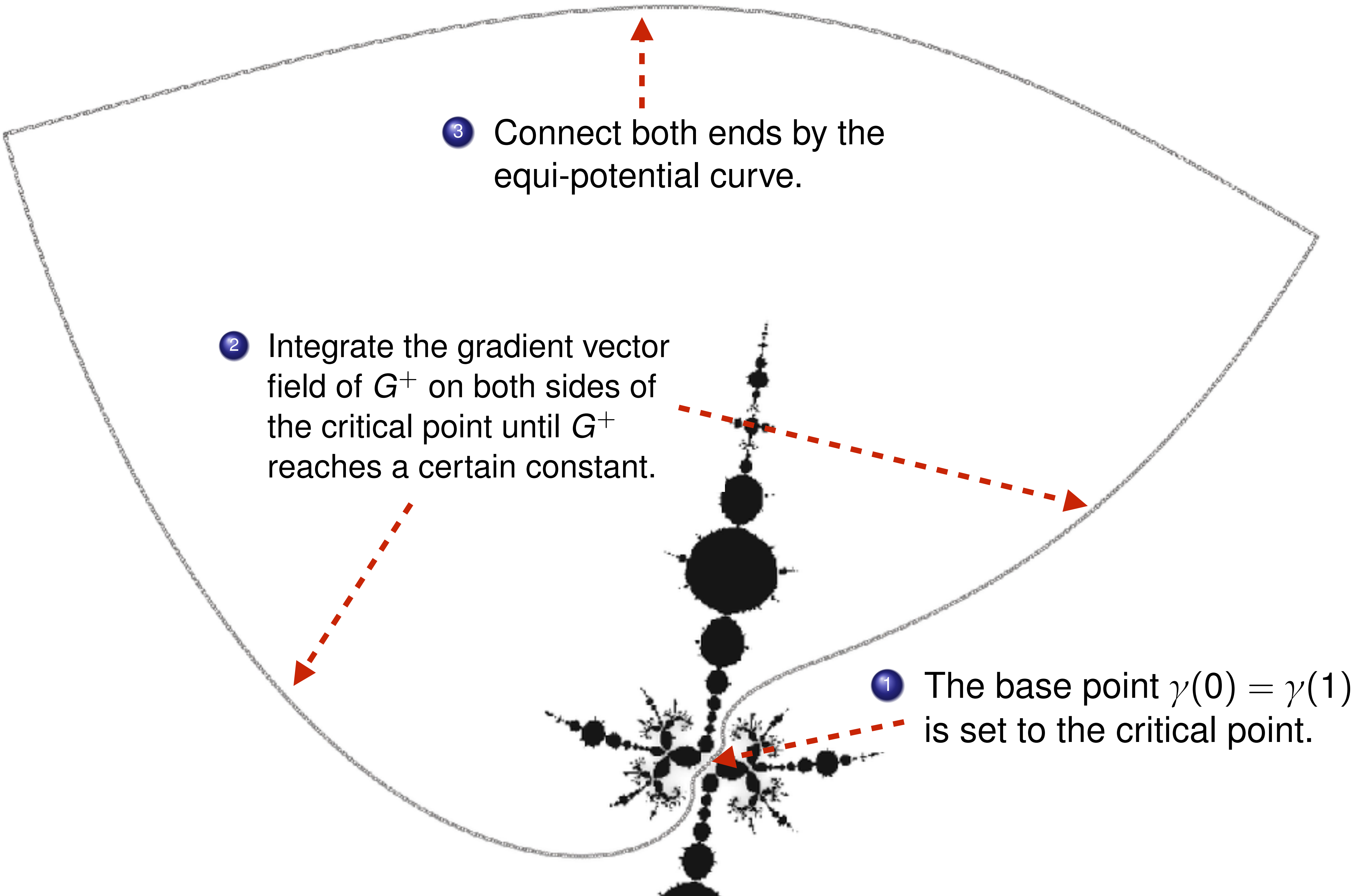
$$G^+(\phi(z_0)) < C.$$

Then $\phi(\gamma)$ should enclose a point of K_f.

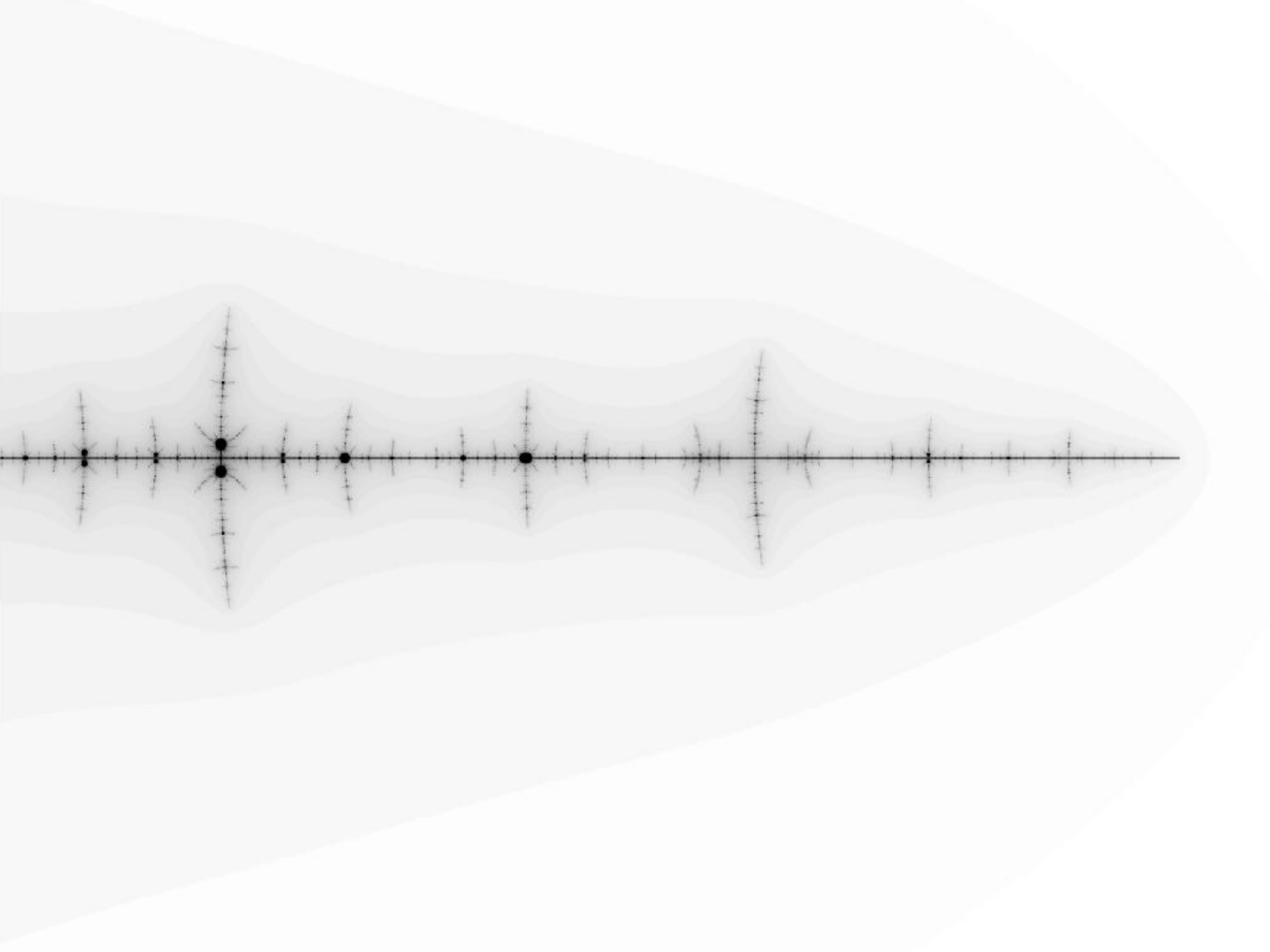
Theorem

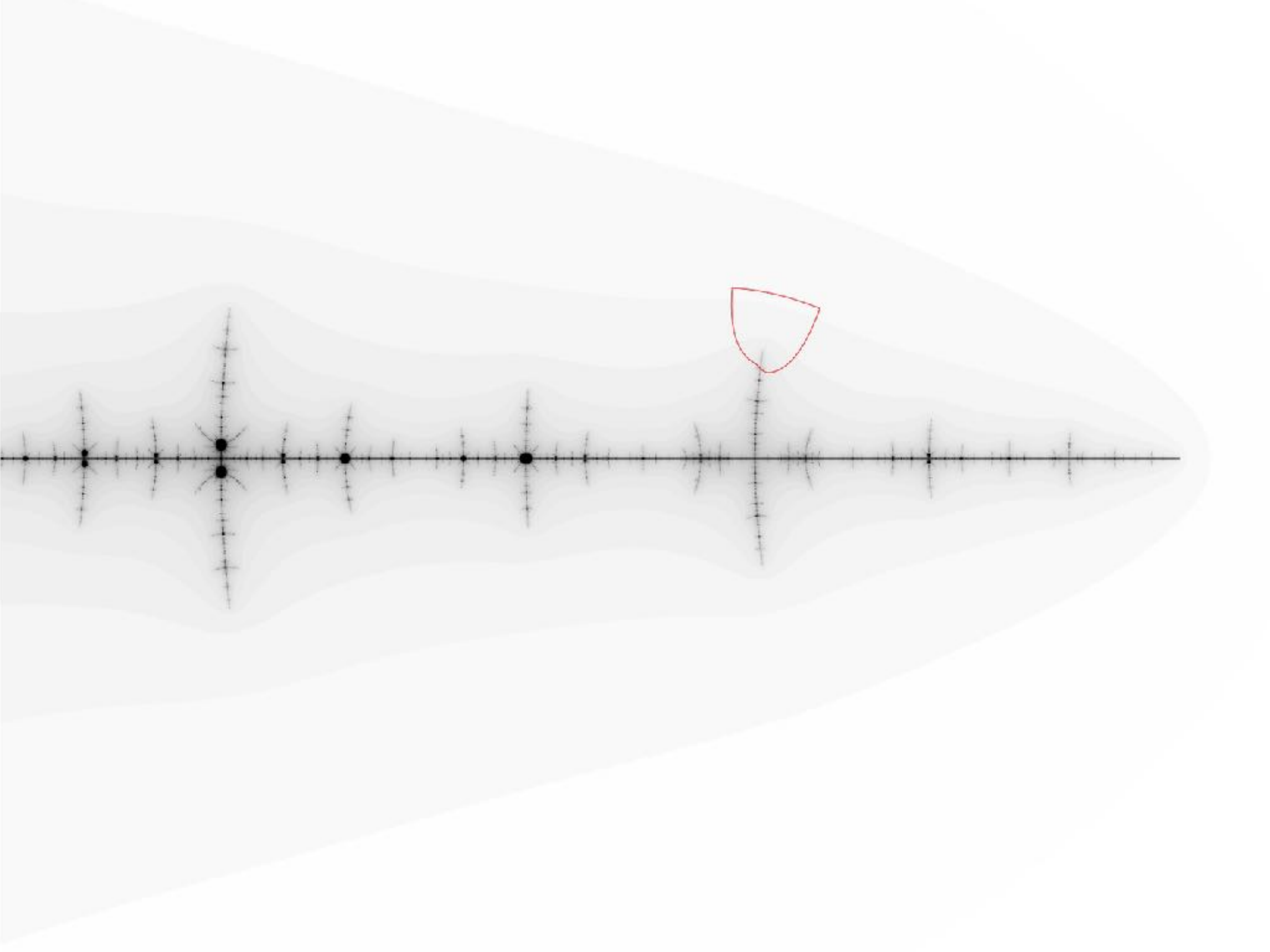
If there is a loop that satisfies the assumption of the Lemma and does not enclose the origin of $\mathbb{C}$, then J_f is disconnected.

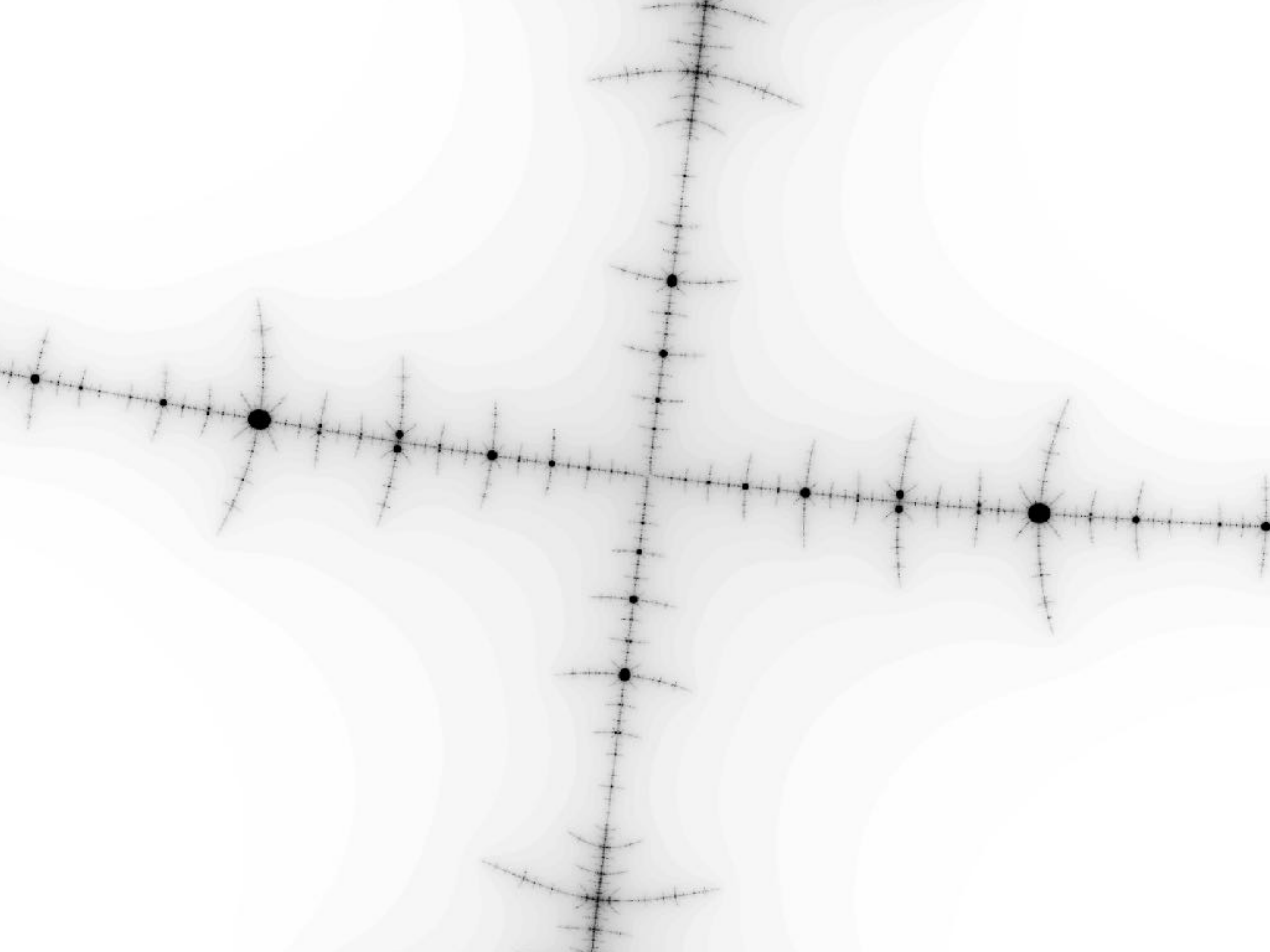
Steps to construct the loop

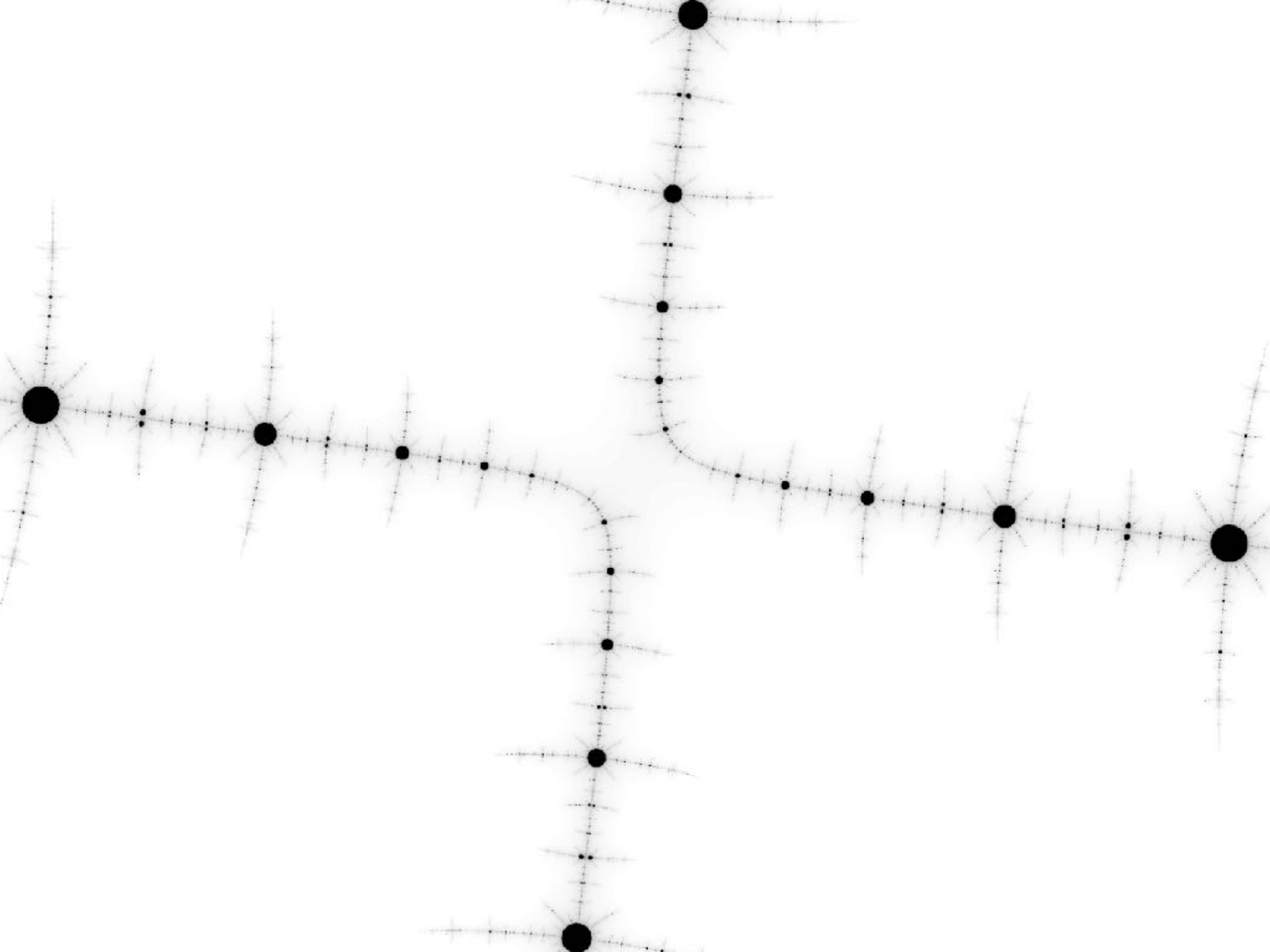


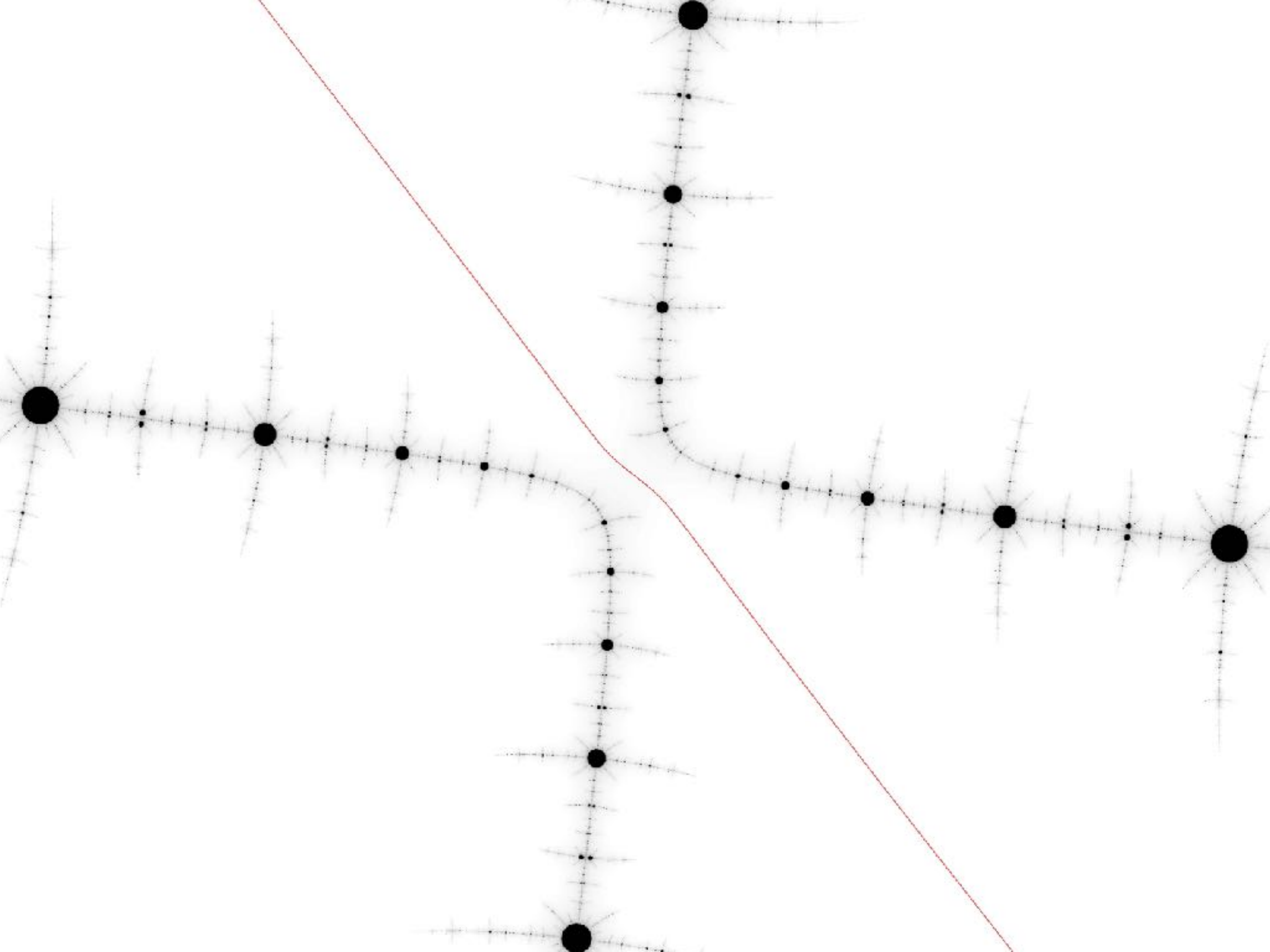
Examples











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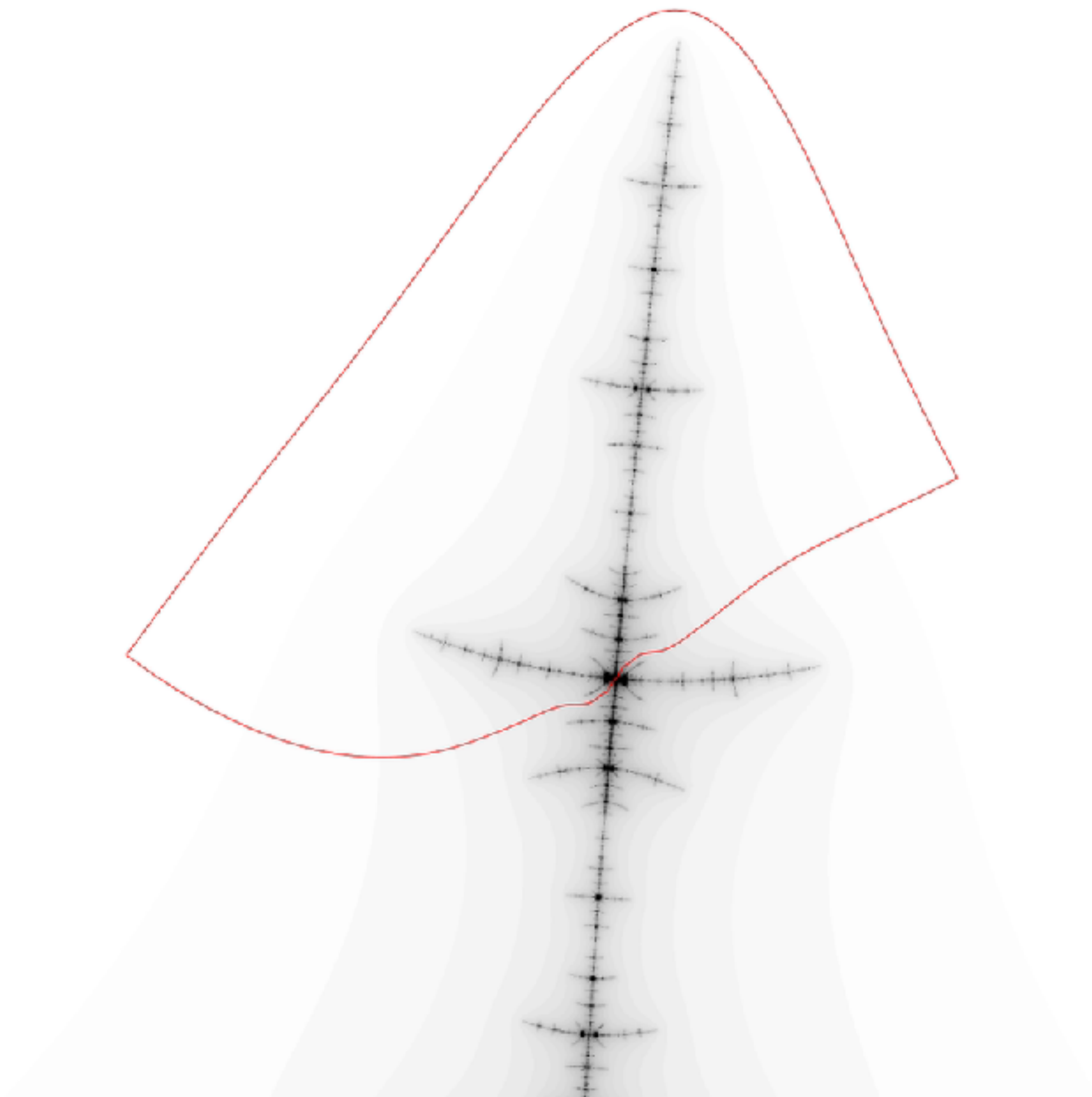
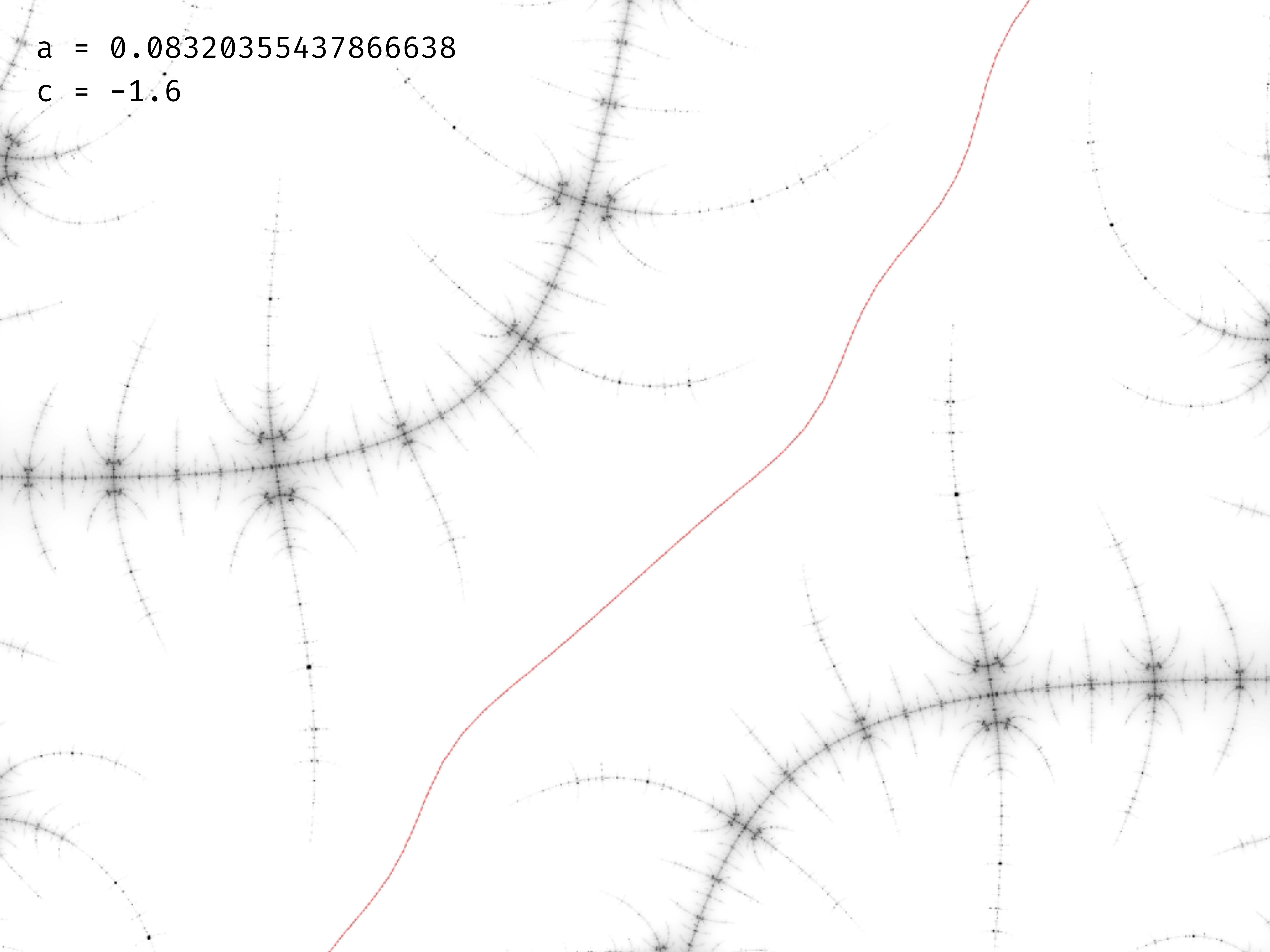
c = -1.6



a = 0.0832035543786638
c = -1.6



Theorems and a conjecture

Joint work with Yutaka Ishii (Kyushu University)

Theorem (connectedness): The Hénon map with (a, c) = (0.23, -1.93) is hyperbolic and non-planner with a connected Julia set.

Let $\mathcal{M} = \{(a,c) \in \mathbb{C}^2 \mid f_{a,c} \text{ has a connected Julia set}\}$

Theorem: $\mathcal{M} \cap \mathbb{R}^2$ is not connected.

Conjecture: $\mathcal{M}$ is not connected.